Dell PowerFlex 5.1.x and PowerFlex Manager 5.1.x

Technical Overview

October 2026
Rev. 1.2

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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Contents

Chapter 1: Introduction **6**
Common PowerFlex terms 6

Chapter 2: Revision history **7**

Chapter 3: PowerFlex overview **8**
New features 8
PowerFlex erasure coding 10
PowerFlex consumption models 11
Data confidentiality using Data at Rest Encryption (D@RE) 12

Chapter 4: Product limits and system requirements **13**
Product limits 13
System requirements 16
Software packages 16
PowerFlex management platform resource requirements 16
PowerFlex cluster component requirements 16
Physical server requirements for storage nodes 17
Physical server requirements for compute nodes 19
Additional hardware considerations 20
Supported hypervisors and operating systems 20
Required Linux packages 20
Microsoft Windows support 21
Browser support 21

Chapter 5: Architecture **22**
MDM cluster 22
Virtual IP addresses for the MDM cluster 23
System stability 23
Storage nodes 23
Storage components 24
Protection domains 24
Device groups 24
Storage pools 25
Over-provisioning control 25
Write cache 25
Volumes 25
Storage data clients and NVMe hosts 26
Storage data clients 28
NVMe over TCP 30
Volume copies 34
Snapshots 34
Thin clone 35
Volume families 35

Snapshot policies....36
Performance....37
Protection and load balancing....37
Spare capacity....37
Rebuild....38
Rebuild throttling....38
Rebalance....39
Rebalance throttling....40
Checksum protection....40

Chapter 6: PowerFlex management platform....41
Capabilities....41
PowerFlex Manager....42
Serviceability....43
Deployment options for each PowerFlex consumption model....43
Dedicated management cluster model....44
Integrated PowerFlex management (non-KVM)....45
Customer-provided infrastructure for PowerFlex management....46

Chapter 7: Networking....48
Network requirements....48
Network limitations....49
Network considerations....49
IP address configurations....49
Network limitations when using VMware ESXi....50
Network stability....50

Chapter 8: Implementing PowerFlex....52
Physical layer....52
Deploying MDMs in a multirack system....53
Approved RAID controller....53
SAN virtualization layer....53
Audit logging....54
Implementation considerations....54
Configuring volumes....54
Naming PowerFlex objects....55
Consumption models....55
PowerFlex software....55
PowerFlex appliance....55
PowerFlex rack....56
PowerFlex custom node....57
Dell PowerFlex for Public Cloud....57
PowerFlex implementation in a VMware ESXi-based system....57
Components in a PowerFlex VMware ESXi-based implementation....58
Supported VMware vSphere features....59
VMware ESXi vStorage APIs for Array Integration....59
VMware vSphere Storage I/O Control....59
Configuring direct attached storage....60

Chapter 9: Maintenance.....61
Maintenance mode 61
Maintain the physical layer..... 61
Maintaining the virtualization layer.....61
PowerFlex failures and recovery cases.....62
Monitoring.....65
SDC connection monitoring.....65
SMART hardware monitoring.....66

1

Introduction

This document provides a high-level technical overview of PowerFlex. It describes the PowerFlex erasure coding protection schemes, the key PowerFlex features, and the management capabilities of the PowerFlex management platform. This document also contains information regarding the different PowerFlex offerings.

The target audience for this document includes customers, presales, and Dell Technologies Support.

This Dell Technologies product uses an embedded Linux distribution that is licensed to Dell. This embedded Linux is only authorized to be used to operate the Dell product and only the Dell product.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to Dell PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

2

Revision history

Table 2. Revisions

Date	Document revision	Description of changes
October 2026	1.2	Physical server requirements for storage nodes and editorial updates
August 2026	1.1	Editorial updates
June 2026	1.0	Initial release

3

PowerFlex overview

PowerFlex is an enterprise-class, software-defined block storage solution that is deployed, managed, and supported as a single system. PowerFlex is hardware-agnostic, flexible, and scalable for enhanced storage capacity and performance. PowerFlex can be deployed either on premises or in AWS and Microsoft Azure multicloud environments.

PowerFlex provides a storage platform and architecture based on erasure coding, a method of data protection that provides improved data durability, resiliency and reliability, and higher capacity usage that reduces storage overhead. PowerFlex uses dual parity erasure coding, which is implemented using multiple protection schemes based on the number of PowerFlex storage nodes.

New and existing customers can benefit from the following features of PowerFlex 5.x:

Improved storage capacity

- Up to 192 TB of node raw capacity.
- Increased capacity utilization—up to 80 percent of raw capacity, depending on the protection scheme.

Improved write caching

PowerFlex can use persistent memory (PMEM), including non-volatile dual in-line memory modules (NVDIMMs) and software-defined persistent memory (SDPM), for write caching and metadata persistence within a protection domain. Using PMEM for write caching improves performance and reduces storage device wear.

NOTE: PowerFlex no longer uses PMEM for inline data compression.

Improved data protection

- Up to 1,022 snapshots per volume family (formerly vTree).
- Data is protected during maintenance and rebuilds.
- PowerFlex can sustain up to two concurrent node or device failures.

Flexible scaling

With PowerFlex, resources such as storage and compute can be scaled together or separately, nondisruptively, and in small or large increments. The system can scale from a few nodes to hundreds of node in a cluster, linearly scaling I/O performance and throughput.

New features

This PowerFlex release introduces the following new features:

Table 3. New features

Feature	Details
Allocation granularity of 1 GB	The minimum size of a volume is 1 GB. In this release, you can increase the volume size in increments of 1 GB. For example, if you have a volume that is 10 GB, you can increase it to 11 GB.

Table 3. New features (continued)

Feature	Details
Boot from SAN	In this release, PowerFlex supports booting from a Storage Area Network (SAN), which is a feature that enables servers to boot an operating system installed on external SAN-based storage, rather than a local disk.
CloudLink	Added support for CloudLink 8.3. CloudLink 8.3 and later versions are supported only on PowerFlex platforms. **NOTE:** CloudLink 8.2 The Key Management Interoperability Protocol (KMIP) server and machine encryption functionalities are available. CloudLink 8.3 KMIP server and machine encryption functionalities remain operational but are deprecated. They are no longer covered under standard support and are planned to be disabled in a future release. Dell Technologies strongly recommends planning a transition to supported alternative solutions to ensure continued functionality and support. For guidance on available options or migration planning, contact your Dell Technologies account representative.
Dynamic licensing	In this release, PowerFlex supports dynamic licensing. Dynamic Licensing offers an enhanced licensing experience by automating license management activities, such as registrations, capacity expansions, and subscription renewals. You can register software instances and share licensing information between the product instance and Dell. Dynamic licensing introduces the ability to manage license allocations more easily across product registrations, enabling a flexible and keyless experience. Through PowerFlex Manager, you can register your device and migrate from key-based XML format licensing to dynamic licensing. See the *Dell PowerFlex Family Licensing Guide* for more information.
Enterprise SONiC support	In this release, PowerFlex rack supports using Dell Networking Enterprise Software for Open Networking in the Cloud (SONiC) for partial network automation in access aggregation switches.
Modify the properties of a storage node (for erasure coding only)	In this release of PowerFlex Manager, you can modify the properties of a storage node. You can complete the following: • Rename a storage node • Add or remove an IP address from a storage node • Modify the role associated with an IP address of a storage node
Pause inventory jobs	In this release, PowerFlex Manager introduces the option to pause scheduled inventory jobs for up to 24 hours, ensuring upgrades can complete without interruption.
PowerEdge 17th generation compute-only nodes	PowerFlex supports using the PowerEdge R670, R770, R6725, and R7725 as compute-only nodes.
PowerEdge 17th generation storage-only nodes	PowerFlex supports using the PowerEdge R670 and R770 as storage-only nodes.
Protection schemes	Added the 4+2 protection scheme.
Storage data clients on VMware Cloud Foundation platform	You can install and upgrade storage data clients on a VMware Cloud Foundation (VCF)-based node or a VMware ESXi 9.x-based node.
User experience enhancements	In this release, PowerFlex Manager delivers a simplified user experience that is related to storage pools, volumes, and snapshots and enhanced capacity metrics reporting.

Table 3. New features (continued)

Feature	Details
	• You can view the security and expiration that is associated with a volume snapshot. • The following capacity metrics are now available through the PowerFlex Manager dashboard: ○ Reducible DRR: The data reduction ratio for the reducible data within the system or protection domain. This metric ignores any unreducible data in the calculation. ○ Unreducible data: The amount of data that is written within the system or protection domain which cannot be reduced by compression or the pattern detection logic PowerFlex uses.

PowerFlex erasure coding

Erasure coding is a data protection method that splits data into multiple fragments, or data strips, encodes it, and distributes the strips across different storage nodes in multiple locations. Erasure coding then creates additional fragments, or parity strips, that are stored on separate parity nodes.

The parity strips are calculated using a formula derived from the original data strips. Parity strips can be used to recover and reconstruct data, even if some fragments or codes are lost or corrupted. This adds redundancy for fault tolerance and data recovery in cases of multiple disk or node failures.

PowerFlex 5.x uses dual-parity erasure coding. Dual parity provides a data protection layout of N strips + 2 parity strips. PowerFlex stores each strip on a separate storage node. This allows for two disk or two node failures within a PowerFlex protection domain before any data unavailability occurs.

The disaggregated compute and capacity in erasure coding-based PowerFlex systems ensures the optimal utilization of storage and compute resources.

The PowerFlex erasure coding architecture supports the following protection schemes:

- 2+2: This protection scheme requires a minimum of five nodes and provides 50 percent usable storage capacity.
- 4+2: This protection scheme requires a minimum of seven nodes and provides 66 percent usable storage capacity.
- 8+2: This protection scheme requires a minimum of 11 nodes. It provides 80 percent usable storage capacity.

To allow for immediate rebuild during a device or node failure, the system is configured with enough spare capacity to rebuild a full node. The spare capacity is configured as a spare node at the level of the device group. You can optionally configure the system to reserve capacity sufficient to rebuild two full nodes.

Data layout

In this architecture, each strip is stored in a different failure domain, which can be either a PMEM or storage device, or a storage node. Any two failed strips can be reconstructed using parity strips.

A strip is the size of the data written to a single device. Each device has a strip capacity of 64 KiB.

An erasure coding stripe is a group of blocks that are spread over multiple devices and built for redundancy. Each stripe spans N + 2 devices (N data blocks residing on D devices + 2 parity strips on two separate devices) and has a usable capacity of 512 KB and another 128 KB of parity (overhead).

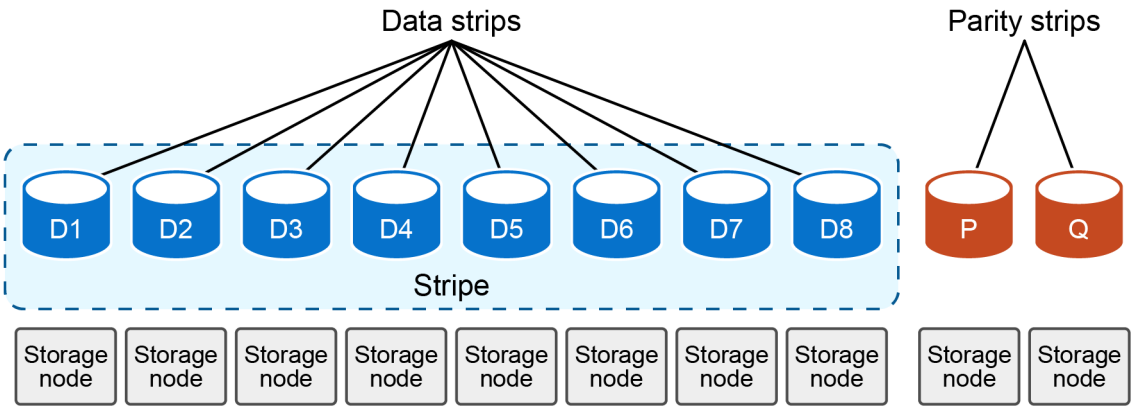


Figure 1. Data layout for 8+2 erasure coding protection scheme

Supported failure examples

The following list explains the types of concurrent disk or node failures supported in an erasure coding environment without incurring data unavailability. In this list, Node A and Node B can be any storage node in the protection domain.

- One or more disks in Node A
- One or more disks in Node A + one or more disks in Node B
- One or more disks in Node A + Node B
- Node A + Node B

PowerFlex consumption models

The PowerFlex family offers several different consumption models. All models are based on PowerFlex block storage software.

PowerFlex software

PowerFlex block storage software deployed on customer-supplied hardware that meets PowerFlex hardware requirements. This implementation is intended for large enterprises where robust onsite hardware support is readily available.

PowerFlex rack

A fully engineered, factory deployed rack-scale system with PowerFlex storage deployed on Dell PowerEdge servers and integrated networking. Delivered ready to power on, PowerFlex rack features a large starting point for PowerFlex that scales to hundreds of nodes and racks.

PowerFlex appliance

A high-performance hardware infrastructure based on Dell PowerEdge servers, with flexible networking options. Requiring configuration and deployment from the customer, PowerFlex appliance provides a smaller starting point than PowerFlex rack, but scales to hundreds of nodes.

PowerFlex custom node

A do-it-yourself networking and management solution in which PowerFlex is installed on Dell PowerEdge servers that have been optimized for PowerFlex systems. PowerFlex custom node allows for broader networking options than PowerFlex appliance.

NOTE: Depending on the configuration of your PowerFlex rack, PowerFlex appliance, or PowerFlex custom node deployment, PowerFlex supports PowerEdge R670, R770, R6725, R7725, R660, R760, R860, R6625, R7625, R650, R750, R6525, and R7525 servers. The servers come optimized for PowerFlex.

Dell PowerFlex for Public Cloud

A deployment of PowerFlex in the public cloud, PowerFlex for Public Cloud is available for both AWS and Microsoft Azure, allowing the same benefits of enterprise-class storage services in the cloud as with on-premises PowerFlex deployments. It provides higher performance, larger volume sizes, and improved resilience over what is currently available on the public cloud.

Dell PowerFlex for AWS	PowerFlex software deployed on EC2 instances in the AWS cloud. PowerFlex for AWS can be deployed on elastic block store (EBS) volumes, or native NVMe SSD drives attached to EC2 instances.
Dell PowerFlex for Microsoft Azure	PowerFlex software deployed on Microsoft Azure virtual machines in the Azure cloud. Based on the use case, PowerFlex for Microsoft Azure can be deployed on Azure managed disks or virtual machines with attached NVMe SSDs .

NOTE: PowerFlex for AWS and PowerFlex for Microsoft Azure are only supported on specific models of AWS EC2 instances and Microsoft Azure virtual machines, respectively.

PowerFlex management platform

All PowerFlex consumption models include the PowerFlex management platform, the management and orchestration software stack for PowerFlex. The PowerFlex management platform includes PowerFlex Manager, which is a web user interface that provides ease of access to PowerFlex management platform functionality. You can also access the PowerFlex management platform through the PowerFlex REST API or the PowerFlex CLI (SCLI).

Data confidentiality using Data at Rest Encryption (D@RE)

Confidentiality is one of the key pillars of the confidentiality, integrity, and availability (CIA) security triad. PowerFlex offers data confidentiality through Dell Data at Rest Encryption (D@RE).

PowerFlex supports Dell CloudLink D@RE-based encryption. CloudLink offers the following encryption methods:

CloudLink software encryption	Dell CloudLink can be integrated with PowerFlex to provide software-based encryption for PowerFlex storage nodes that is transparent to the features and operation of the PowerFlex solution. CloudLink uses dm-crypt, a native Linux encryption package, to secure storage node devices. A proven high-performance volume encryption solution, dm-crypt is widely implemented for Linux machines.
Hardware-based FIPS 140-2 compliant data at rest encryption	For hardware-based D@RE, you can select from a supported list of self-encrypting drives (SEDs) that meet your business needs, using integrated CloudLink for key management.

CloudLink encrypts the storage node devices with unique keys that are controlled by enterprise security administrators. CloudLink Center provides centralized, policy-based management for these keys, enabling single-screen security monitoring and management across one or more PowerFlex deployments.

4

Product limits and system requirements

PowerFlex product limits specify the minimum and maximum supported configurations in this release. The maximum and minimum values for key system parameters are dictated by the product limits.

Product limits

The following table summarizes the supported capabilities and system limits.

NOTE: The supported limits described in the following table might differ from the enforced system limits listed in the output for the `scli --query_system_limits` command.

Table 4. Supported capabilities and system limits

PowerFlex capability	PowerFlex software limits	PowerFlex appliance and PowerFlex rack limits
Device size	Minimum: 480 GB Maximum: 16 TB	Minimum: 3.84 TB Maximum: 16 TB
Maximum acceleration devices of type SDPM or NVDIMM (maximum that is supported in a PowerFlex node)	PowerEdge R840/R740xd/R640 nodes: • NVDIMM: 6 PowerEdge R750/R650 nodes: • NVDIMM: 8 PowerEdge R760/R660 nodes: • SDPM: 2	PowerEdge R840/R740xd/R640 nodes: • NVDIMM: 6 PowerEdge R750/R650 nodes: • NVDIMM: 8 PowerEdge R760/R660 nodes: • SDPM: 2
Maximum data networks for MDM virtual IPs	4	4
Maximum data networks per system	8	4
Maximum device groups per protection domain	1 storage device group + 1 PMEM device group	1 storage device group + 1 PMEM device group
Maximum devices per protection domain	4096	768
Maximum erasure coding compression ratio	10x the system's raw capacity. The system is defined as all storage devices in the protection domains.	10x the system's raw capacity. The system is defined as all storage devices in the protection domains.
Maximum hosts (SDC or NVMe) per system	SDC: 2,048 NVMe: 1,024 + 1,024 SDCs	Managed hosts: 768 (SDC or NVMe or both, on either storage nodes or compute-only nodes) Unmanaged hosts: 2,048
Maximum logical capacity per system	4 EB (Base 2)	4 EB (Base 2)
Maximum managed PowerFlex nodes per system	N/A	400
Maximum number of concurrent logged-in management clients (PowerFlex Manager, REST API, and SCLI)	128	128

Table 4. Supported capabilities and system limits (continued)

PowerFlex capability	PowerFlex software limits	PowerFlex appliance and PowerFlex rack limits
Maximum number of configured syslog servers	16	16
Maximum number of snapshots in a single volume family	1022	1022
Maximum number of snapshots per snapshot policy	1022	1022
Maximum number of volumes-to-SDC or NVMe host mappings per system	SDC: 262,143 NVMe: 32,000	SDC: 262,143 NVMe: 32,000
Maximum persistent memory (PMEM) devices per storage node	NVDIMM: 2 SDPM: 2	NVDIMM: 2 SDPM: 2
Maximum PowerFlex nodes per resource group	N/A	32
Maximum protection domains per system	8	8
Maximum raw capacity per protection domain	16 PB	16 PB
Maximum raw capacity per storage node	307 TB	307 TB
Maximum snapshot policies per system	1000	1000
Maximum storage devices (drives) per storage node	32	32
Maximum storage nodes per system	128	128 The total number of storage-only and compute-only managed nodes cannot exceed 400.
Maximum system raw capacity	8+2: 16 PB 4+2: 4 PB 2+2: 4 PB	8+2: 16 PB 4+2: 4 PB 2+2: 4 PB
Maximum total number of volumes and snapshots in the system. Can be a single protection domain or storage pool.	128,000	128,000
Maximum total number of volumes and snapshots in a protection domain	32,768	32,768
Maximum user accounts	256	256
Maximum volumes that can be mapped concurrently using the REST API (PowerFlex Manager, REST API, and SCLI)	1024	1024
Maximum volumes that can be mapped to a single SDC	1024	1024
Minimum capacity per node	1.6 TB	1.6 TB x 5 (8 TB)
Minimum devices per node	1	5
Minimum number of storage nodes in a protection domain	8+2: 11 storage nodes 4+2: 7 storage nodes	8+2: 11 storage nodes 4+2: 7 storage nodes

Table 4. Supported capabilities and system limits (continued)

PowerFlex capability	PowerFlex software limits	PowerFlex appliance and PowerFlex rack limits
	2+2: 5 storage nodes	2+2: 5 storage nodes
Minimum storage pool/device group size	1.6 TB x number of storage nodes in the protection domain	5 disks x 1.92 TB x minimum node count
Number of storage pools per protection domain	1	1
System over-provisioning factor	100x size of the storage pool	100x size of the storage pool
SCSI-3 reservation type	Write exclusive registrants only	Write exclusive registrants only
Total size of all volumes per storage pool	4 EB (Base 2)	4 EB (Base 2)
Volume size	Minimum 1 GB Maximum 8 PB (Base 2)	Minimum 1 GB Maximum 8 PB (Base 2)

Table 5. NVMe over TCP limits

PowerFlex item	Limit
Maximum SDTs per system	512
Maximum SDTs per protection domain	128
Minimum SDTs per protection domain	2
Maximum number of NVMe hosts created in PowerFlex	1024
Maximum NVMe host connections (I/O controllers) per SDT	512
Maximum NVMe host connections (I/O controllers) per system	65,519
Maximum I/O controller queue depth	128
Maximum I/O controller queues	32
Number of volumes mapped to a single NVMe host	1024
Maximum number of paths per volume	8
Maximum number of connections per NVMe host per protection domain	128

Table 6. PowerFlex management platform limits

Capability	Limit
Maximum number of configured users	256
Maximum number of LDAP users	256
Maximum number of LDAP servers	3
Maximum number of DNS servers	3
Maximum number of time sources	3
Maximum number of Syslog servers	3
Maximum number of SMTP servers	3
Maximum number of vCenters	6

Related information

SAN virtualization layer

System requirements

This section lists the requirements for PowerFlex system components and the minimum PowerFlex Manager resource requirements for each consumption model.

Software packages

Depending on the type of deployment, PowerFlex requires different software packages.

See the release notes for your PowerFlex offering for a full of list of required software or firmware. Release notes are available at Dell PowerFlex Family: Product Documentation.

PowerFlex management platform resource requirements

The following section lists the requirements for system components and the minimum PowerFlex management platform resource requirements.

The following table summarizes the PowerFlex management platform storage and resource requirements for PowerFlex software depending on your PowerFlex management cluster cluster scale.

Table 7. PowerFlex software storage and resource requirements

System scale	CPU cores	Storage capacity (GB)	RAM (GB)
Low Scale	8	300	24
Medium Scale	8	350	26
High Scale	14	450	32

The following table summarizes the PowerFlex management platform storage and resource requirements for PowerFlex rack and PowerFlex appliance.

Table 8. PowerFlex rack and PowerFlex appliance storage and resource requirements

CPU cores	Storage capacity (GB)	RAM (GB)
14	450	32

PowerFlex cluster component requirements

Each PowerFlex cluster includes the following components. Before deploying PowerFlex, ensure that the PowerFlex environment meets the server requirements for the cluster type.

Storage nodes

The number of required storage nodes depends on the erasure coding protection scheme being used:

- 2+2 protection scheme: A minimum of five storage nodes
- 4+2 protection scheme: A minimum of seven storage nodes
- 8+2 protection scheme: A minimum of 11 storage nodes

Metadata manager cluster

The are five metadata manager (MDM) nodes:

- One primary MDM
- Two secondary MDMs
- Two tiebreakers

For PowerFlex software and PowerFlex appliance, these nodes can be hosted on the same servers as storage nodes or on a separate server. For PowerFlex rack, the MDM nodes must be hosted on separate servers. For PowerFlex for AWS, and PowerFlex for Microsoft Azure, the MDM nodes must be hosted on separate AWS EC2 instances or Azure virtual machines.

PowerFlex management platform options

Table 9. PowerFlex management platform options

PowerFlex offering	PowerFlex management platform options
PowerFlex rack	A dedicated PowerFlex management cluster (KVM) of three nodes.
PowerFlex appliance	One of the following options: • Dedicated PowerFlex management cluster (KVM) of three nodes. • Single ESXi PowerFlex management node. • Customer-provided infrastructure for PowerFlex management (ESXi). • Customer-provided infrastructure for PowerFlex management (KVM/Linux). • Customer-provided infrastructure for PowerFlex management (KVM/Linux) bare metal (available only by special request).
PowerFlex custom node	One node as a dedicated PowerFlex management cluster. **NOTE:** In the single-node configuration, a hypervisor is required, such as KVM or VMware ESXi, to host the PowerFlex management platform virtual machines.
PowerFlex custom node or PowerFlex software	Integrated PowerFlex management (non-KVM) PowerFlex management cluster. or Customer-provided infrastructure for PowerFlex management (KVM/Linux) or ESXi environment to host the PowerFlex management cluster.

Additional PowerFlex components

Additional PowerFlex components include:

- Storage data clients (SDCs) up to the maximum number supported, deployed on separate compute nodes.
- NVMe hosts up to the maximum number supported. The NVMe hosts co-exist on the same hosts as the storage nodes.

Physical server requirements for storage nodes

Servers used as storage nodes must meet the requirements described below.

PowerFlex offers a sizer tool to make the complex work of sizing the physical requirement. Go to the PowerSizer and review the physical sizing considerations on the page. Dell Technologies recommends using the sizer tool for all and not perform manual sizing.

Initial deployment must use matching node configurations. Do not mix nodes with differing hardware configurations in the initial erasure coding deployment. Although all nodes should have the same server generation, number of disks, and disk capacity, the processor core count, memory size, and persistent memory capacity must be identical across all nodes. Include as many matching nodes as possible during the initial deployment to support the 8+2 protection scheme for optimal capacity efficiency, as this selection cannot be changed later.

After a valid initial deployment, supported higher specification nodes can be added during expansion, but performance will be limited by the lowest-specification nodes in the configuration.

The following tables summarize the physical requirements for Dell PowerEdge R660, R760 , R860, R650, and R750 servers.

Table 10. Storage node physical server requirements

Component	Requirement
Total CPU requirements	2*16 cores with hyperthreading enabled

Table 10. Storage node physical server requirements (continued)

Component	Requirement
Physical memory (RAM)	256 GB - 512 GB 16 cores CPU needs to be paried with 256 GB. 32 cores CPU needs to be paired with 512 GB. **NOTE:** Ensure that SWAP is disabled. PowerFlex does not support using SWAP on storage nodes. Using SWAP with PowerFlex slows the performance of the system and can slow down the I/Os on the system disks that are used for SWAP. It may also impact the following: ● MDM repository-related operations ● PowerFlex logs written to disk. For instructions on disabling SWAP, see How to Permanently Disable Swap Partition in Linux.
Disk space	● Disk space formula per node: 1 GB * number of nodes in the system + 1 GB (for log collection) ○ Required path locations: `/opt` ○ 1 GB for log collection under `/tmp` (reused every time a log bundle is created)
Connectivity	● 2 * 25/100 GbE network ● Dual-port network interface cards Ensure the following: ● There is network connectivity between all components. ● Network bandwidth and latency between all nodes is acceptable, according to application demands. ● The Ethernet switch supports the bandwidth between network nodes. ● MTU settings are consistent across all servers and switches. ● The TCP ports are not used by any other application, and are open in the local firewall of the server: **NOTE:** You can change the default ports.

Table 11. Storage node operating system and boot disk requirements

Item	Value
Disk type	BOSS M2 SSD/PCIe SSD
RAID	2*M.2 devices in RAID 1
Minimum capacity	960 GB
Endurance	1 DWPD over 5 years or 200 PB life span writes
Sustained performance IOPS (random 4 KB)	Read: 120,000 IOPS
	Write: 30,000 IOPS
Sustained performance bandwidth (sequential)	Read: 2,800 MB/sec
	Write: 150 MB/sec
NOTE: Due to a limitation, do not install any of the following on the same operating system disk as the MDM: PowerFlex Manager and PowerFlex installer.	

Physical server requirements for compute nodes

Servers used as compute nodes must meet the requirements described in the following tables.

Table 12. Compute node physical server requirements

Component	Requirement
Processor	Minimum: 2*16 cores with 256 GB RAM
CPU threads	• NVMe initiator: Default 2; max 8 • SDC: Default 2; maximum 8 • LIA: 1
Physical memory (RAM)	• NVMe target initiator: 6 GiB • SDC: 6 GiB
Disk space	• Disk space formula per node: 1 GB * number of nodes in the system + 1 GB (for log collection) ○ Required path locations: `/opt` ○ 1 GB for log collection under `/tmp` (reused every time a log bundle is created) • For SDC running on Linux or Citrix Hypervisor (XenServer): 1 GB **CAUTION: Only current customers who have installed XenServer or Citrix Hypervisor are supported, new customers are not advised to install XenServer or Citrix Hypervisor.** **NOTE:** The minimum operating system size for a three-node cluster is 10 GB. • For VMware ESXi : 80 GB is the minimum for a boot disk for VMware topologies; the PowerFlex storage VM resides on the boot datastore.
Connectivity	• 2 * 25/100 GbE network • Dual-port network interface cards Ensure the following: • There is network connectivity between all components. • Network bandwidth and latency between all nodes is acceptable, according to application demands. • The Ethernet switch supports the bandwidth between network nodes. • MTU settings are consistent across all servers and switches. • The TCP ports are not used by any other application, and are open in the local firewall of the server: **NOTE:** You can change the default ports.

Table 13. Compute node operating system and boot disk requirements

Item	Value
Disk type	BOSS M2 SSD/PCIe SSD
RAID	2*M.2 devices in RAID 1
Minimum capacity	960 GB
Endurance	0.5 DWPD over 5 years or 200 PB life span writes
Sustained performance IOPS (random 4 KB)	Read: 120,000 IOPS
	Write: 30,000 IOPS
Sustained performance bandwidth (sequential)	Read: 2,800 MB/sec
	Write: 150 MB/sec

Additional hardware considerations

If you are purchasing a Dell PowerEdge-based solution in addition to a PowerFlex software license, ensure that you select hardware and components that are identical to the hardware and components supported for PowerFlex engineered solutions (PowerFlex rack, PowerFlex appliance, or PowerFlex custom node).

PowerEdge configurations that differ from configurations supported by PowerFlex engineered solution are at risk of encountering issues. Aligning to an engineered solution is critically important when selecting drives for a storage device.

- Agnostic drives should be avoided. Vendor-specific drives must be selected to ensure that the PowerFlex nodes ship with PowerFlex-validated drives.
- To ensure PowerFlex software compatibility, Dell Technologies recommends that you configure the solution using a PowerFlex product base (PowerFlex rack, PowerFlex appliance, or PowerFlex custom node).

Supported hypervisors and operating systems

PowerFlex storage nodes and compute-only nodes support using several hypervisors and operating systems.

For the most up-to-date list of supported operating systems, see the Dell PowerFlex Simple Support Matrix for this release.

PowerFlex storage nodes support the following:

- PowerFlex embedded operating system
- VMware ESXi 9.x for NVMe over TCP in unmanaged mode
- VMware ESXi 8.0 for NVMe over TCP clients

PowerFlex storage data client (SDC) compute-only nodes support the following:

- VMware ESXi 9.x in unmanaged mode
- VMware ESXi 8.0

The latest patches of VMware ESXi and vCenter for compute-only nodes are tested periodically. VMware has discontinued support for specific CPUs. See CPU Support Deprecation and Discontinuation in VCF Releases for details.

Required Linux packages

The following table lists the packages that must be installed on each PowerFlex node running Red Hat Enterprise Linux or SUSE Linux Enterprise Server (SLES) operating systems.

Table 14. Required packages

PowerFlex component/node	Requirement
All components for both Red Hat Enterprise Linux and SLES	• numactl • libaio1 • wget • apr • libapr1 • python2-rpm or python3-rpm • yum-utils
SDC	• which
MDM components	• bash-completion (for SCLI completion) • Latest version of Python 3.x for newer Linux distributions • binutils • java-17-openjdk-headless
Storage nodes (PDS and DGWT)	• smartmontools • smartmontoolslibs • binutils • sg3_utils • hdparm • pciutils

Table 14. Required packages (continued)

PowerFlex component/node	Requirement
Diagnostic tools	• sar • sysstat
PowerFlex-supported storage controllers	PERCCLI
Storage node components that contain NVDIMMs	• ndctl • jq • daxio • libpmem
Secure authentication mode	Ensure that OpenSSL 64-bit v1.1 is installed on all servers in the system: • `libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm` • `openssl1-1.0.1g-0.40.1.x86_64.rpm`
LDAP	Ensure that OpenLDAP 2.4 is installed on all servers.
LIA	pciutils
NVMe initiator	Linux nvme_tcp and nvme_fabrics modules must be loaded in the operating system. Use `lsmod | grep nvme` to validate that they are available.

Microsoft Windows support

Microsoft Windows is supported only on the PowerFlex storage data client (SDC) and LIA components.

To use secure authentication mode, ensure that the following are installed on all servers in the system:

- OpenSSL 64-bit v1.0.1 or later (v1.1, however, is not supported)
- Visual C++ redistributable 2010 package, 64-bit

The Microsoft Windows Storage Spaces feature (available from Windows 8.1 and later) is not supported in Microsoft failover clusters running on PowerFlex volumes.

Browser support

PowerFlex supports Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

5

Architecture

This section presents the main components of the PowerFlex 5.x architecture, which is based on erasure coding.

MDM cluster

The metadata manager (MDM) is the monitoring and configuration agent for PowerFlex. It is the authority that controls and tracks data storage ownership, mapping, and protection, and manages rebuilds and all system-related functions. No I/Os run through the MDM.

The MDM is involved in:

- Capacity allocation
- Spare allocation
- Uber health/ degrade writes
- Rebuild and/rebalance

To support high availability, a cluster of five metadata managers are installed on different PowerFlex nodes: one primary MDM, two secondary MDMs, and two tiebreakers. One MDM is assigned the primary MDM role, and the others act as secondary or tiebreaker MDMs:

- The primary MDM controls the storage nodes, SDCs, and storage data targets (also known as the NVMe targets). The primary MDM also contains and updates the MDM repository, the database that stores the storage node configuration, and how data is distributed between the nodes in the system. This repository is constantly replicated to the secondary MDMs, so they can take over control of the repository with no delay. Every MDM cluster has one primary MDM.
- The secondary MDMs are ready to take over the primary MDM role, if necessary. The two secondary MDMs in the cluster allow for two points of failure, increasing resiliency.
- The sole role of the tiebreaker MDMs is to help determine which MDM is the primary MDM in the event of MDM failover. There are two tiebreakers in the cluster. This ensures that there are always an odd number of MDMs in a cluster, which guarantees that there is always a majority in electing the primary MDM.

In addition, the secondary MDMs are designated "replica" MDMs—meaning, each one contains a copy of the primary MDM repository. The replica allows a nearly instantaneous switchover to a secondary MDM in the event of primary MDM failure.

Each PowerFlex deployment may contain a standby MDM, which is not a part of the cluster. In the event of MDM node failure, the standby MDM can be promoted to a cluster role. You can also prepare standby managers and tiebreaker nodes, for a total of thirteen cluster and standby MDMs.

The MDM cluster and standby MDMs are limited to 16 IP addresses, which include all cluster members (primary, secondary, standby MDM, and standby tiebreaker).

The same MDM package is installed on all members of the MDM cluster, which is created automatically when deploying a system with any of the PowerFlex automated deployment tools. MDMs do not require dedicated nodes. They can be installed on nodes hosting other PowerFlex components.

The following figure illustrates a typical five-node MDM cluster:

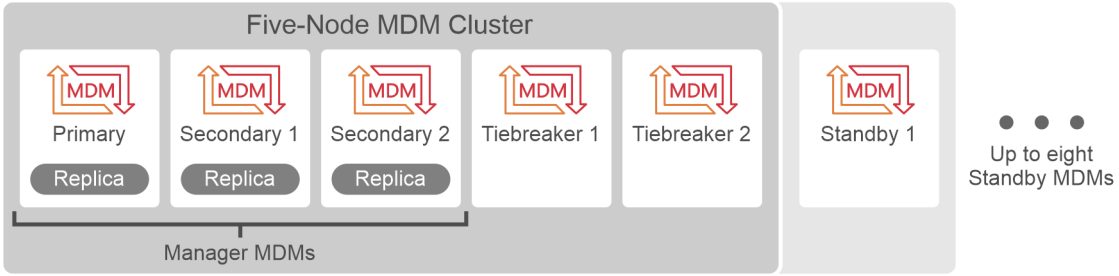


Figure 2. Five-node MDM cluster

Virtual IP addresses for the MDM cluster

You can define up to four virtual IP addresses for the MDM cluster. SDCs are then mapped to the virtual IP addresses, instead of to the static MDM IP addresses. MDMs are sometimes switched during normal operation of the cluster, and the virtual IP address will always be mapped to the active MDM.

The use of virtual IP addresses simplifies maintenance procedures on the MDM cluster, because system components communicate using the virtual IP addresses. Therefore, SDCs do not need to be reconfigured when a server hosting an MDM is replaced.

In new installations, you can add and map the virtual IP addresses for the MDM cluster using the CSV installation file. The REST API can also be used to add virtual IP addresses to the cluster. In all cases, a virtual IP NIC placeholder must be mapped to each virtual IP address. Ensure that NICs are available for this purpose.

Existing systems may be extended to include additional MDMs in a cluster. The new MDMs should be mapped to the existing virtual IP addresses.

If virtual IP addresses must be modified, you must use the PowerFlex CLI (SCLI) or the REST API, exercising extreme caution. All SDCs then require reconfiguration to reflect the changes made to the MDM cluster. Otherwise, the SDCs cannot communicate with the MDM cluster, and volumes are not accessible.

System stability

The MDM proxy ability augments the network stability capabilities.

In the PowerFlex architecture, each storage data client (SDC) is required to be fully connected to the MDM for control and management of actions. If an SDC is unable to reach the MDM, the SDC moves to disconnected state, which at some point could lead to complete I/O failure.

If a network failure occurs, PowerFlex allows clusters that do not use the MDM virtual IP address capability to use the secondary MDMs as a proxy for all control traffic to provide greater redundancy.

If a network failure occurs and the MDM proxy is in use, the management API raises an appropriate alert.

Storage nodes

Storage nodes are physical or virtual servers that run your chosen PowerFlex operating system and store PowerFlex information. Storage nodes support the actions that affect software components running on the operating system and the storage and PMEM devices that are installed on the physical node.

You can add a storage node to your PowerFlex system to scale up your storage without having to add compute resources. You can distribute data across multiple storage nodes over a fault-tolerant, high-speed network. If one storage node fails, the PowerFlex system can continue operating with minimal impact and continue to protect against data loss. Storage nodes are accessed and managed under the **Lifecycle** tab in PowerFlex Manager.

Each storage node is installed on a server that contributes storage devices and optional persistent memory (PMEM), or an AWS EC2 instance or Microsoft Azure virtual machine that contributes storage devices only. Storage nodes contain the following components:

- PowerFlex data server (PDS): The frontend application that is responsible for processing host I/Os.
- Device gateway target (DGWT): The component that is accountable for placing data on the storage drives using erasure coding protection schemes.

The minimum number of storage nodes that are deployed in the cluster depends on the erasure coding protection scheme that you are using:

- 2+2 protection scheme: five storage nodes.
- 4+2 protection scheme: seven nodes
- 8+2 protection scheme: 11 storage nodes.

Storage nodes can be temporarily removed from live systems for maintenance and troubleshooting purposes, but cannot be permanently removed.

Storage components

The PowerFlex architecture includes the following storage elements:

- Protection domains
- Device groups
- Storage pools

Together, these elements link the physical PowerFlex layer to the virtual storage layer.

Protection domains

A protection domain is a logical entity that consists of a group of storage nodes that back each other up.

Each storage node belongs to one protection domain, so each protection domain contains a unique set of storage nodes that are defined to behave as an independent system. All storage objects (volumes, storage pools, and the like) are contained within the protection domain.

There are no logical connections between nodes in different protection domains.

The following image displays a PowerFlex deployment with three protection domains. Protection domain 1 is configured in an 8+2 protection scheme, and protections domains 2 and 3 are configured in a 2+2 protection scheme. The protection domains all share the same MDM cluster (configured on protection domain 1), as required for a single deployment.

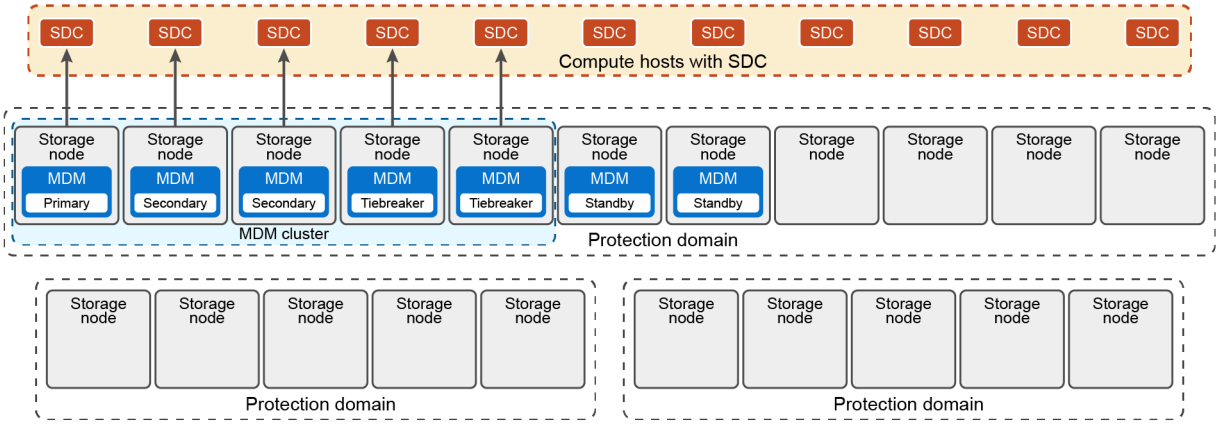


Figure 3. Multiple protection domains in a single PowerFlex deployment

Each protection domain supports up to 128 storage nodes. This enables:

- Optimal performance
- A reduction of theoretical *mean time between failure* issues
- The ability to sustain multiple failures in different protection domains

Protection domains are configured exclusively to support the designated protection scheme. You can add protection domains during installation and modify existing protection domains after installation.

Device groups

A device group is a set of physical devices within a protection domain. Device groups represent the raw capacity of a group of devices. A device group is responsible for grouping together devices of a similar type, which makes device management easier. Device groups can also be used for tiering.

PowerFlex supports two different types of device groups: storage devices (SDPMs) and persistent memory devices (NVDIMMs). Each device group must contain storage devices on a minimum of four separate nodes.

Device groups containing storage devices only support SSD and PCI SSD (NVMe) drive types. Dell Technologies recommends that only one drive type be configured for each device group for optimal performance.

Device groups are physically bound by the amount of physical storage in each group. Device groups must be uniform in terms of number of devices, with the same number of devices in each device group. You can view device groups under the **Block** tab in PowerFlex Manager.

NOTE: Dell Technologies does not recommend mixing different types of SSDs and NVMe drives in a device group. PowerFlex does not perform optimally if there are differences in size or type.

Storage pools

A storage pool is a group of volumes. A storage pool allocates capacity from a device group to create the volumes, and sets the base erasure coding protection scheme, which is either 2+2, 4+2, or 8+2. Only one storage pool is configured for each device group and its protection domain.

The protection scheme that is selected during deployment cannot be changed to an alternate protection scheme after deployment. For example, customers starting with a 2+2 (five-node) protection scheme cannot move workloads to a 4+2 (seven-node) or an 8+2 (11-node) protection scheme. Restriping is not supported. Restriping is the process of changing the geometry, such as from 2+2 to 8+2.

Storage pools can be created from nodes with different device types. The mixing of SAS, SATA, and NVMe is permitted, but mixing device types reduces overall storage performance to that of the slowest device. Individual SDSs must have drives of a single type, but mixing those nodes together is allowed.

PowerFlex enables data compression by default, but it can be disabled during storage pool creation. You can also optionally define the over provisioning ratio for the storage pool.

Over-provisioning control

Over-provisioning is an optional control that allows you to set a limit that is defined as a percentage factor of the physical size of the storage pool. For example, if the storage pool is 10 TB and the over-provisioning factor is 200%, you can create thin volumes up to 20 TB. The default over-provisioning value is unlimited.

When over-provisioning is set, the system enforces the over-provisioning limit. If the sum total of the sizes of all volumes and snapshots created for the storage pool reaches the over-provisioning limit, no new volumes or snapshots can be created.

Write cache

Write cache (WrC) is a feature that provides high availability and low write latency, and aggregates writes to data stored in persistent memory such as non-volatile dual in-line memory modules (NVDIMMs) and software-defined persistent memory (SDPM), or to storage devices such as SSDs or NVMe drives.

The write cache is a distributed component that provides high availability by writing to storage on multiple nodes. All writes and write updates are first written into the write cache, which can be either a persistent memory device or subsystem or a storage device.

When persistent memory is configured in the system and enabled on the storage pool, it is used for write caching and provides better performance than using storage devices. Persistent memory also reduces the number of writes to storage devices, and uses batteries as redundant power sources.

NOTE: Persistent memory is not used for inline data compression, which is enabled by default in the erasure coding schema.

Volumes

Volumes are a portion of the storage pool capacity, evenly distributed across storage devices. They appear as a single object in PowerFlex.

Volumes must be mapped to hosts before an application or operating system can access them.

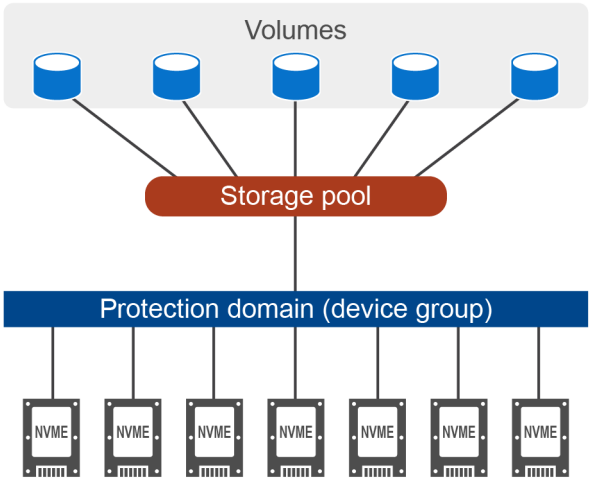


Figure 4. Volumes within the PowerFlex architecture

Only thin volumes are supported in the PowerFlex erasure coding architecture. Capacity is “on reserve,” but not allocated until used. This policy enables more flexibility in provisioning. Whereas thick capacity is limited to available capacity, thin capacity provisioning can be oversubscribed, as follows:

- Maximum thin capacity provisioning = 5 * (total raw capacity - used capacity)
- When capacity usage reaches the level where it may cause I/O errors, alerts are generated. At certain higher capacity levels, volumes can no longer be created.

Define volume names according to PowerFlex object naming conventions.

Related information

Naming PowerFlex objects

Storage data clients and NVMe hosts

PowerFlex supports both storage data clients (SDCs) and NVMe hosts to map and present volumes to the operating system. SDCs and NVMe hosts can co-exist in the same cluster; the only restriction is that they cannot use the same volume simultaneously.

The SDC is a PowerFlex-proprietary lightweight device driver that provides frontend volume access to applications, and maps and presents volumes to the operating system or hypervisor. SDCs maintain peer-to-peer connections to every PDS managing a pool of storage. They translate SCSI to PowerFlex protocol.

NVMe hosts consume PowerFlex block storage using NVMe over Fabrics (NVMe-oF) connectivity to map and present volumes to the operating system or hypervisor. From a user perspective, NVMe hosts play the same role in the system as SDCs. However, they co-exist on the PowerFlex storage nodes and do not require any additional hardware.

The following figure displays the PowerFlex processes using NVMe and SDC hosts.

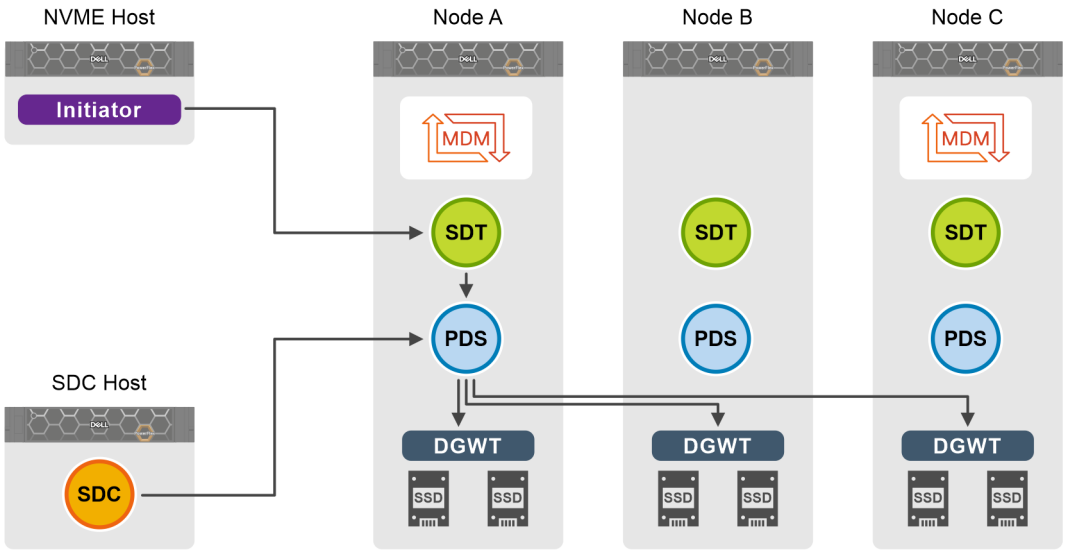


Figure 5. PowerFlex processes and services

The following figures show the difference in connectivity between the SDC and the NVMe host. The SDC sits on the application host or compute node and connects directly to the storage layer. The NVMe host, on the other hand, connects to an NVMe target called the storage data target, which run on the storage side. The storage data target directs the volumes onward to the storage components.

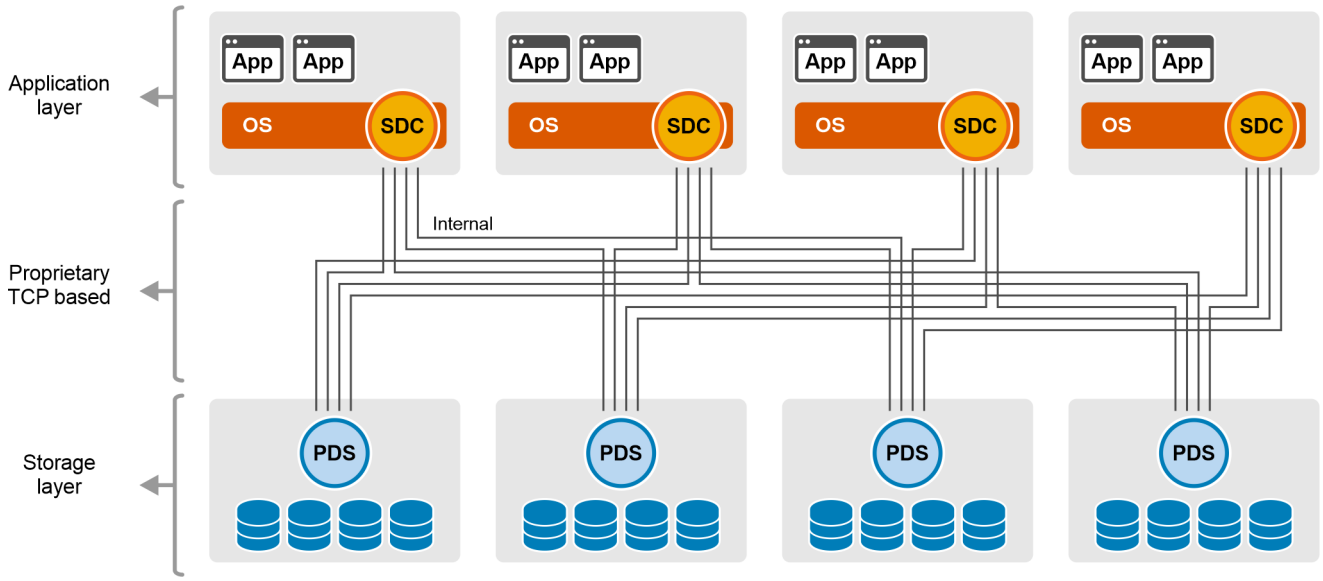


Figure 6. SDC host connectivity

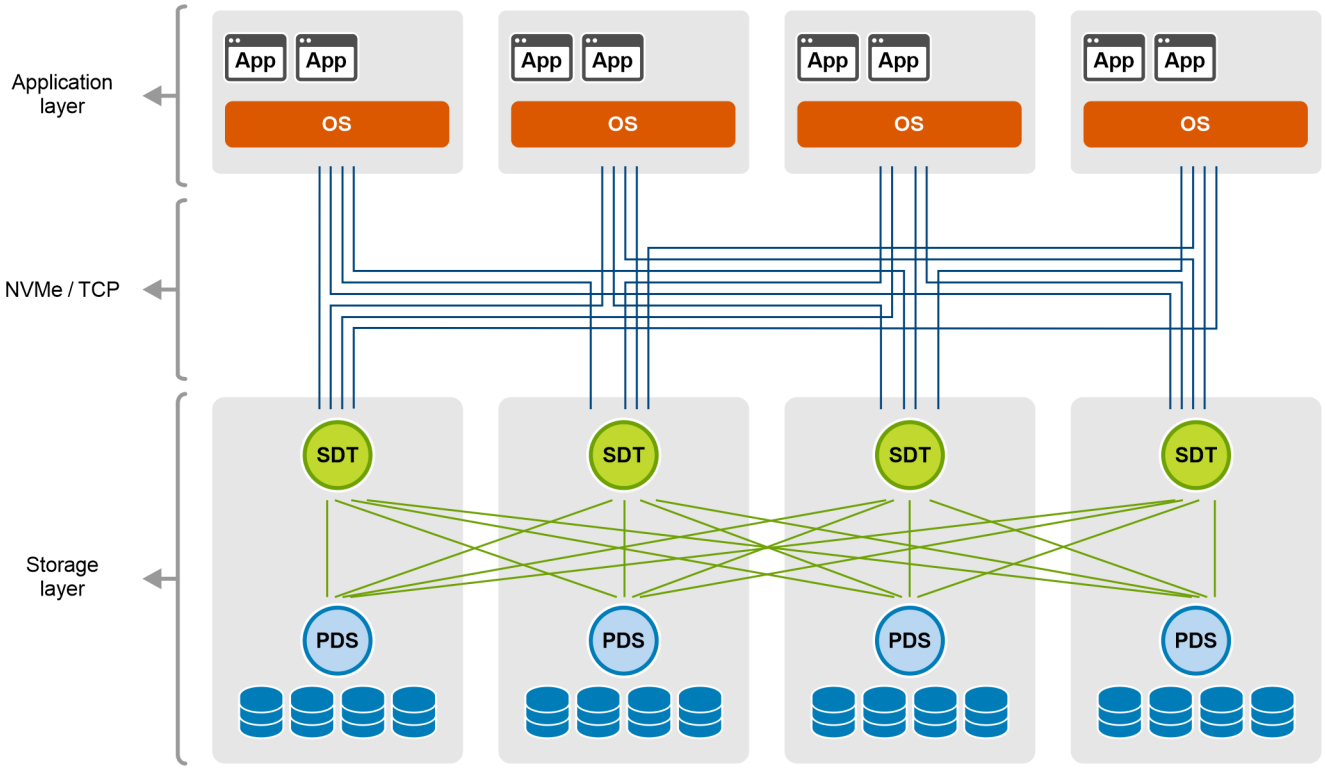


Figure 7. NVMe over TCP host connectivity

Storage data clients

The storage data client is a driver installed on hosts that translates the PowerFlex control and data path protocols to the SCSI layer. The SDC handles all the PowerFlex multipathing aspects and connects to all the PowerFlex storage components.

The SDC also integrates with relevant operating system logic to provide a native block storage device in the operating system. Volumes (also know as LUNs in other storage systems) are mapped to the SDC to be consumed as block devices.

SDCs are supported on Linux, Windows, AIX, and other operating systems, and can be installed on any non-PowerFlex host meeting PowerFlex specifications for SDCs.

SCSI support

The SCSI identity consists of the following:

- The VMware ESXi SDC registers a new SCSI adapter with Fibre Channel transport.
- The Fibre Channel node worldwide name is the low 64 bits of the SDC GUID.
- The Fibre Channel port worldwide name is the high 64 bits of the SDC GUID.
- The Fibre Channel port and node worldwide names are also exposed for the virtual Fibre Channel targets.

Every mapped volume contains the following:

- A SCSI worldwide name which can be queried at a vital product data (VPD) page `0x83` EUI-64 designator using the SCSI Inquiry command. The EUI-64 designator is 128 bits, and it is composed of the PowerFlex system MDM ID and Volume ID.
- A SCSI unit serial number can be queried though a VPD page `0x83` EUI-64 designator using the SCSI Inquiry command. For example:

```
MOS_SNPRINTF((char *)&pUnitSerial->serialNum[0],
serialNumBufSize, "%s-%.16"MOS_X64_BIT__STRING"-%.16"MOS_X64_BIT__STRING,
MOS_VERSION__LEGACY_COMPANY_NAME_CAP, mapVol_GetMdmID(pVol), mapVol_GetVolID(pVol));
```

  The serial number follows the pattern `EMC - MDM_ID-Vol_ID`.

Every mapped volume is assigned a Target LUN address when it is mapped. Because the storage I/O (SIO) volume ID is 64 bits, the address is calculated as follows:

- The SDC exposes up to 8 K LUNs in total. Begin with the volume index (low 32 bits of the volume ID) and use the modulo operation with 8 K. The result is the SCSI volume index.
- The low eight bits of the SCSI volume index become the LUN ID, and the remaining top five bits becomes the target ID. If there is a collision, which is when a volume exists with a TargetLun pair, the LUN ID stays a low eight bits and the target ID increments by one until the collision is resolved. If a collision exists for all 32 target IDs, the LUN ID increments by one and starts over until the collision is resolved.
- SDC reports a single target worldwide name that is based on the SDC GUID. This target ID is unrelated to the target that is part of the volume LUN ID.

Supported Vital Product Data pages

PowerFlex supports the following pages for SCSI Vital Product Data (VPD):

- `0x00` - Standard inquiry
- `0x80` - Unit serial number
- `0x83` - Device identification
- `0x80` - Block limits
  - This page includes size limitations used by the optional commands: Unmap, Write Same, and Compare and Write.
- `0x83` - Logical block provisioning

Supported SCSI commands

The following tables describe the SCSI commands that PowerFlex supports.

Table 15. Mandatory SCSI commands

Command type	Operation code	Comments
FORMAT UNIT	04h	N/A
INQUIRY	12h	INQUIRY commands support a subset of the standard pages.
READ	READ (10): 28h READ (12): A8h READ (16): 88h READ (32): 7Fh/0009h	PowerFlex supports various sizes of READ commands: READ (10), READ (12), READ (16), and READ (32).
READ CAPACITY (10)	25h	N/A
READ CAPACITY (16)	9Eh/10h	N/A
REPORT LUNS	A0h	N/A
TEST UNIT READY (TUR)	00h	N/A

Table 16. Optional SCSI commands

Command type	Operation code	Comments
COMPARE AND WRITE	89h	Atomic test and set (ATS)
MODE SENSE - only page `0xa` / subpage `0x0`	1Ah	MODE SENSE commands support a subset of the standard pages.
Reservations related		
PERSISTENT RESERVE IN	5Eh	SCSI-3 reservation support
PERSISTENT RESERVE OUT	5Fh	SCSI-3 reservation support'
RESERVE	0x16	SCSI-2 reservation support. PowerFlex supports RESERVE 1 and RESERVE 6.

Table 16. Optional SCSI commands (continued)

Command type	Operation code	Comments
RELEASE	0x17	SCSI-2 reservation support. PowerFlex supports RELEASE 6 and RELEASE 10.
UNMAP	42h	Thin provisioning or block delete
WRITE	WRITE (10): 2Ah WRITE (12): AAh WRITE (16): 8Ah WRITE (32): 7Fh	PowerFlex supports different sizes of WRITE commands: WRITE(10), (WRITE 12), WRITE (16), and WRITE (32)
WRITE SAME	WRITE SAME (10): 41h WRITE SAME (16): 93h WRITE SAME (32): 7Fh	Zero block or write zeroes. PowerFlex supports different sizes of WRITE SAME commands: WRITE SAME (10), WRITE SAME (16), WRITE SAME (32)

The following command is not supported:

- EXTENDED COPY

NVMe over TCP

NVMe over TCP is based on NVMe over Fabrics (NVMe-oF), which is an industry-standard protocol for storage transport. NVMe-oF is supported by multiple operating systems and does not require any host agent on the storage node.

TCP provides a standard, common, interoperable solution that delivers excellent performance. The TCP transport allows the use of existing networks and hardware.

PowerFlex hosts can use standard NVMe over TCP operating system drivers to connect to PowerFlex. They do not require proprietary host agents. These drivers are available in mainstream Linux distributions and in VMware ESXi.

NVMe over TCP is supported in a disaggregated, two-layer PowerFlex deployment architecture. The NVMe host is deployed on a compute node and communicates directly with the NVMe target, which is known as the storage data target (SDT), on the storage node. NVMe hosts are deployed through PowerFlex Manager.

Operating system support

PowerFlex supports NVMe over TCP on the following operating systems:

- VMware Cloud Foundation 9.0
- VMware ESXi 9.0
- VMware ESXi 8.0 U3
- SUSE Linux Enterprise Server (SLES) 15 SP6, running on x86_64 architecture with kernel version 5.14.21-150500.55.36-default or higher

For the most recent list, see the NVMe/TCP Host/Storage Interoperability Simple Support Matrix.

Feature support

- PowerFlex supports NVMe reservations (SCSI-3 equivalent) with NVMe over TCP.
- PowerFlex supports Clustered VMDK on VMware ESXi 8.0 U3 and higher.

NVMe hosts

PowerFlex supports connecting and consuming storage from NVMe hosts. An NVMe host is a host running an operating system that supports NVMe over TCP.

After an NVMe host is created in PowerFlex, you can map volumes to the host so that the NVMe host can consume storage. The NVMe host connects to an NVMe target called the storage data target, which runs on the storage side. The storage data target directs the volumes onward to the storage components.

The following NVMe host parameters are configurable:

- Number of connections. For hosts that are expected to generate a high I/O load, you can increase the default value.
- Number of paths per volume. Up to eight paths can be configured per volume in the host attribute. This is mostly an operating system limit per volume and per total path.

PowerFlex supports specifying networks that might fail together. NVMe over TCP connections are allocated to the NVMe over TCP hosts through multiple network sets, which offer resiliency to the failure of a network set.

Storage data target (NVMe target)

The storage data target (SDT) is the channel for communications with NVMe hosts. It serves as the NVMe target and translates NVMe over Fabrics protocol into internal PowerFlex protocol.

When NVMe over TCP is used, the storage data target is deployed on each storage node and provides access to the volumes inside that protection domain.

In PowerFlex, every storage data target provides a discovery service. The information returned to the host includes the ports assigned to the host, either by automatic load balancing or by manual connectivity configuration.

Boot from SAN

PowerFlex supports booting from a storage area network (SAN). Boot from SAN is a feature that enables servers or VMs to boot an operating system installed on external SAN-based storage, rather than booting from locally-attached storage.

Boot from SAN requires a server BIOS and a NIC that support booting from an NVMe over TCP SAN-based boot image, and support NVMe over Fabrics. Supported operating systems include SUSE Linux Enterprise Server 15 and VMware ESXi 9.

Persistent discovery

PowerFlex supports persistent discovery for NVMe over TCP targets. When NVMe hosts are initially discovered through the persistent discovery controller, they remain connected to the discovery service. This enables automatic updates of the hosts regarding storage side changes.

When a change in the discovery information is provided to the NVMe host, the discovery controller returns an asynchronous event notification and the host requests the updated discovery log page.

Changes in discovery information include new volumes mapped to the host from a new protection domain, and new storage data targets added to the system. The discovery log page is also updated when load balancing moves the host connection to a different storage data target.

When using this functionality, confirm that the host operating system supports the persistent discovery controller and that it is enabled for the host. This feature is supported with VMware ESXi 8.0, Red Hat Enterprise Linux 9, and SLES 15 SP6, and later. For more information, see the respective operating system for the NVMe over TCP host configuration.

Load balancing

PowerFlex supports NVMe over TCP connectivity and adds NVMe over Fabrics targets to enable hosts to connect to storage. When hosts are allowed to connect and consume storage, the system ensures that it remains resilient, balanced, and available.

PowerFlex NVMe over TCP load balancer

Utilize the optimal PowerFlex system performance by distributing the NVMe host workloads over the NVMe targets (storage data targets). PowerFlex offers two ways to achieve load distribution: manual and automatic. In both cases, system ports are selected for the host and are returned to the host when it performs NVMe discovery.

Automatic load balancing, which is the default policy, uses an algorithm that selects the system ports based on the following:

- Access to the mapped volumes of the host
- Path resiliency
- Load balancing

The host connectivity plan is recalculated under the following conditions:

- Host connects.
- Storage data target is added or removed
- Network set or system network is modified
- Storage data target is added or removed to a storage system
- Storage data target IP address is reassigned to a different network

NOTE: The load balancer ignores temporary issues such as network failures or if a PowerFlex storage node goes down for a few minutes.

The storage system provides multiple ports through which the host can connect, though every port and node has a limited throughput. To achieve the best performance, ensure that the combined I/O load of the hosts is well distributed over the available system ports and nodes. The NVMe over TCP load balancer aims to automate the planning and execution of the host-to-storage connectivity task to achieve path resiliency and workload balance.

The PowerFlex load balancer considers the host networks for resiliency. When a host loses a network, it can still use other paths through available networks to access the storage.

NVMe over Fabrics discovery

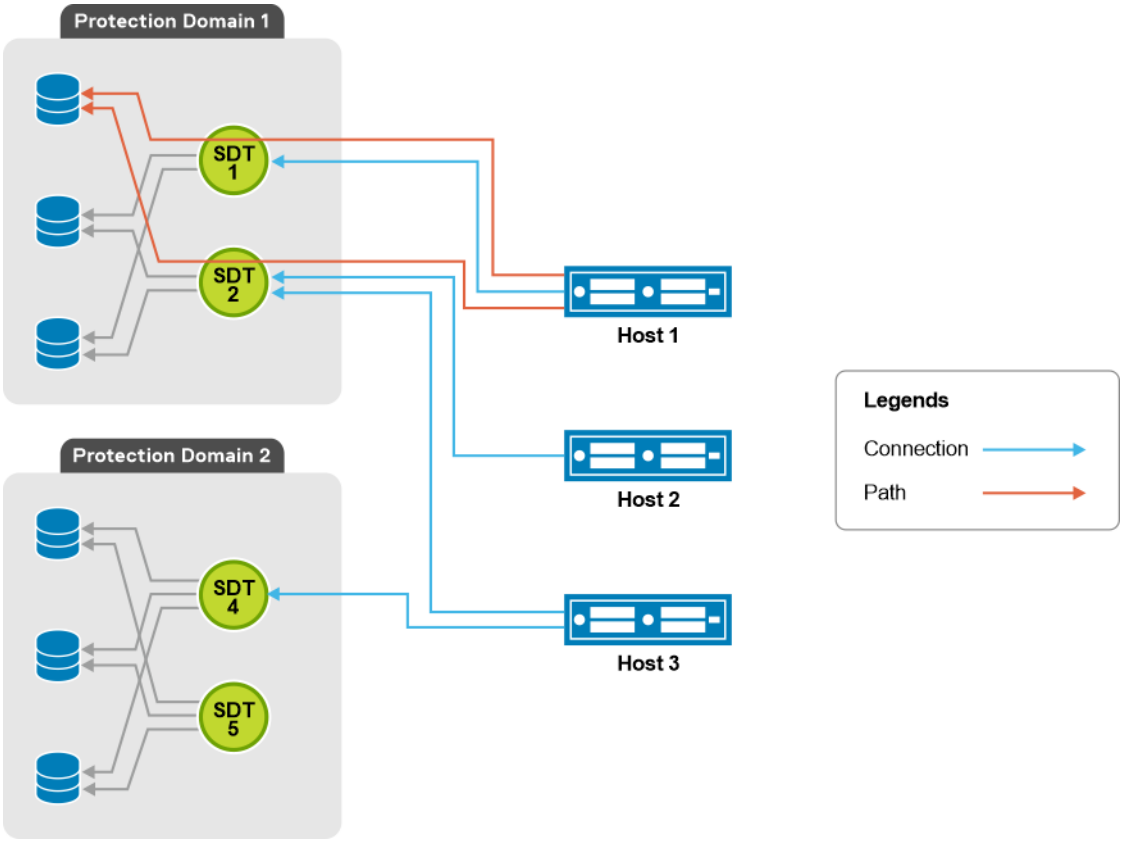
NVMe over Fabrics includes a discovery protocol which is a standard method for the host to discover and connect to available subsystems.

The host connects to a discovery service and receives discovery information, including all the connection ports to access its volumes. PowerFlex continues to update the hosts regarding changes using the persistent discovery controller. In PowerFlex, every storage data target provides a discovery service. The information returned to the host includes the ports that are assigned to that host by the load balancer. The host can also be configured manually If the user selects the manual connectivity configuration.

Balancing host connections

When a host is mapped to the first volume from a protection domain, it assigns ports that might connect to volumes in the protection domain. Each port represents a connection. The number of ports that are assigned is a configurable host attribute, which is up to 128 per protection domain. The system selects the least-occupied ports to ensure fault isolation. The number of paths per volume is a configurable host attribute, with a maximum of eight paths per volume. This configurable host attribute is mostly an operating system limit per volume and total paths.

NVMe over TCP connections and paths



A connection is the establishment of the NVMe controller and includes the following:

- Link between a host part and a specific storage data target IP
- Connection couple (host-port and system-port)
- The load balancer provides 10 ports for each host by default and up to 128 ports.

A path is a connection that is used to provide access to a volume. This connection makes the path a tuple (host-port, system-port, and volume), which must be a subset of the connections.

By default, the load balancer provides four paths per volume to each host, which is selected from the assigned ports.

Manual load balancing

Users might prefer to manually configure host connectivity. Manual configuration is appropriate for special cases, such as when hosts generate a higher-than-expected traffic load.

To configure manually, the user designates in the host configuration the system ports to which the host must connect.

Migrating to NVMe over TCP

PowerFlex provides the option to migrate workloads from SDC to NVMe over TCP on VMware ESXi. The migration can be performed either online or offline.

- Online migration using storage vMotion (VMFS only): The standard way of moving storage is with storage vMotion. Storage vMotion also supports switching protocols by migrating to a new datastore. For more information, see the *PowerFlex and PowerFlex Manager Software Administration Guide*.
- Offline migration (VMFS only): There is an option for converting an existing VMFS datastore from SCSI (SDC) to NVMe over TCP without sending all the data over the network. For more information about the offline migration steps, see SCSI to NVMe VMware VMFS Datastore Offline Migration Steps.

NOTE: There is no standard way to migrate Linux environments, VMware ESXi clusters, and RDMs.

Volume copies

Volume copies are pointer-based objects that provide point-in-time copies of data that are stored in snapshots or thin clones for local data protection. These volume copies provide a method of recovery for data that has been corrupted or accidentally deleted.

PowerFlex uses redirect-on-write technology to copy data from one volume to another. This technology ensures that system space is used optimally and reduces the management burden by never requiring administrators to designate protection space.

Volume copies can be created either manually or automatically within a PowerFlex system. They are considered write-order and crash-consistent.

PowerFlex supports two types of volume copies: snapshots, which are read-only copies, and thin clones, which are read/write copies of the data.

Snapshots

Snapshots are read-only copies of volumes that are used for local data protection. They provide a method of recovery for data that has been corrupted or accidentally deleted.

Snapshots are pointer-based objects that provide point-in-time copies of data that are stored in volumes or thin clones. They can be created either manually or automatically within a PowerFlex system and are considered write-order and crash-consistent.

PowerFlex allows you to take up to 1022 snapshots per volume of existing volumes. The snapshots are thin provisioned and are extremely quick.

Because snapshots are read-only objects and cannot be modified, this allows snapshots to serve as ransomware protection if PowerFlex system data is compromised.

Snapshots cannot be mapped to the hosts. Once a snapshot is generated, it becomes a new, unmapped volume in the system. Volumes can be restored from snapshots, and thin clones (writable snapshots) can be created from them.

You can delete snapshots, but cannot resize them.

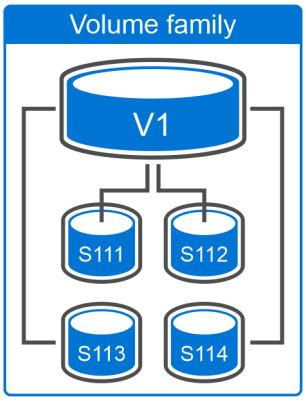


Figure 8. Snapshot operations

Keep the following considerations in mind when using snapshots:

- Because snapshots are not full copies of the original data, they should not be relied upon as a backup or as the disaster recovery solution.
- A write-order, crash-consistent snapshot is not considered application consistent, because the snapshot might not be a full representation of the application dataset at that point in time.
- Typically, a host or client caches data with the intent to write it to the storage resource. Cached data is not available within the storage when a snapshot is taken.
- Snapshots also consume overall system storage capacity to preserve the point in time. Ensure that the storage pool has enough capacity to accommodate snapshots.
- Removing a volume also removes all its direct snapshots.
- Dell Technologies does not recommend trimming volumes that contain snapshots, because the logical capacity may not shrink.

Secure snapshots

The secure snapshots option prohibits deletion of snapshots until the defined expiration period has elapsed.

The secure snapshot setting is for volumes and their thin clones, which is optional when creating a snapshot. When the secure snapshot setting is enabled, the snapshot and its parent object are protected from deletion until the retention period expires. This setting provides a cost-effective line of defense against ransom attacks and accidental deletion of snapshots or volumes.

Thin clone

A thin clone is a read/write copy of a volume or a snapshot that allows you to change its content through writes.

Thin clones are thin copies of the object from which they were created. As with snapshots, thin clones are thin, pointer-based objects that use redirect-on-write technology that provides immediate access to the data contained in the source of the thin clone. Thin clones are not full copies of the original source. Because they share data blocks with the parent resource, they should not be used for disaster recovery scenarios.

The following figure shows an example of a thin clone that is created from a supported resource. When initially created, the thin clone shares all blocks with the resource from which it was created. Due to redirect-on-write technology, as new writes to the original resource or the thin clone are made, new space is consumed, and the original data remains until it is no longer in use.

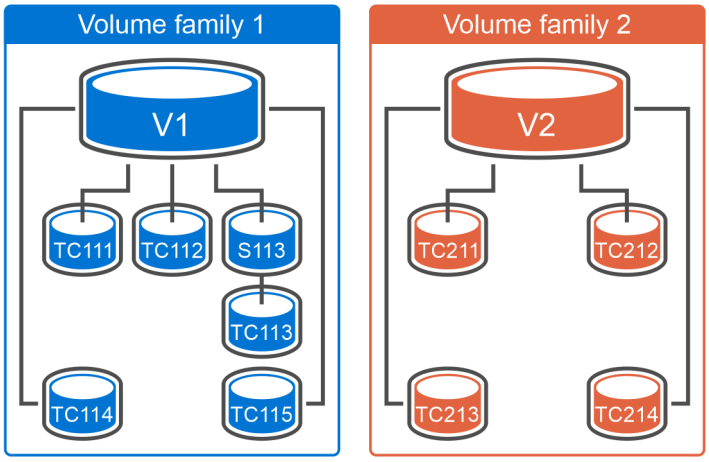


Figure 9. Thin clone operations

Thin clones can be mapped to the host and be used like normal volumes. If you want to map and use the snapshot volume, you must create a thin clone of the snapshot and map it. If you want to delete the parent volume, thin clones are unaffected.

NOTE: If you select **Delete volume with all of its thin clones** during a volume deletion operation, all the thin clones are deleted along with the volume.

Volume families

A volume family consists of the root volume and all its descendant volumes and snapshots.

A volume family is the structure that is comprised of the volume, snapshots, and thin clones resulting from that volume. It is a tree spanning from the source volume at its root, whose descendants are:

- Snapshots of the volume
- Thin clone of the snapshot
- Thin clone of the volume
- Thin clone of the thin clone
- Snapshot of the thin clone

In the following diagram of a volume family:

- TC111 & TC112 are thin clones of V1 (volume1)
- S111 is a snapshot of V1.
- TC211 is a thin clone of snapshot S111.

- S211 is a snapshot of thin clone TC211.
- S311 is a snapshot of thin clone TC112.
- TC311 is a thin clone of snapshot S311.
- TC411 is a thin clone of thin clone TC111.

Together, V1, S1xx, TC1xx, S2xx, TC2xx, S3xx, TC3xx, and TC4xx are the volume family of V1.

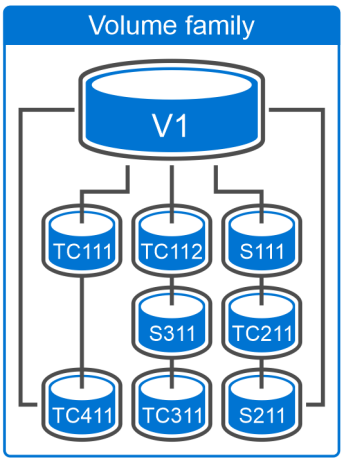


Figure 10. Volume family structure

Snapshot policies

Snapshot policies enable you to define policies where you can configure the number of snapshots to take at a given time for one or more volumes.

Snapshots are taken according to a customer-defined rule. You can define the time interval in between two rounds of snapshots, and the number of snapshots to retain in a multilevel structure. For example, take snapshots every x minutes, hours, days, or weeks. There are one to six levels, with the first level having the most frequent snapshots.

For example, you might define a rule to take snapshots every 60 minutes, with the following retention levels:

- 24 snapshots
- Seven snapshots
- Four snapshots

After the parameters for the policy are defined, the source volume is selected to add to the snapshot policy. You can add multiple source volumes to a snapshot policy, but only a single policy per source volume is allowed. Only one volume per volume family may be used as a source volume of a policy (any policy). When you remove the source volume from the policy, you must choose how to handle snapshots.

You can optionally specify that the snapshots created by the snapshot policy are secured. When snapshots are secured, the snapshots created by the policy cannot be modified or deleted before the end of the snapshot retention period. Also, the parent volume of the snapshots cannot be deleted during the snapshot retention period.

Snapshots that are created by the policy are referred to as auto snapshots. PowerFlex Manager displays the snapshot policy for each snapshot, which also indicates that it is an autosnapshot.

- If the source volume has no auto snapshots, you cannot unassign the source volume from the snapshot policy. However, you can remove auto snapshots in PowerFlex Manager.
- If the source volume has auto snapshots, but none of them are locked, you are prompted to confirm that you would like to delete all auto snapshots. If any of the auto snapshots are locked, the locked auto snapshots are detached from the snapshot policy, but not deleted, and you must remove the auto snapshots manually.

Performance

PowerFlex is a scale-out, high-performance storage system. Adding nodes not only increases capacity, but also increases the performance of the system.

Protection and load balancing

PowerFlex maintains user data in an erasure coding layout. Each piece of data (strip) is stored in a different failure domain (device or node). The copies are distributed over the storage devices according to an algorithm that ensures uniform load of each unit in terms of capacity and expected network load. Rebuild and rebalance processes are fully automated, but also can be controlled manually.

Spare capacity

The spare capacity feature ensures that PowerFlex has enough reserve drive space to enable an immediate rebuild of a full node if a storage node or device fails.

Spare capacity is configured as a spare node at the level of the device group, and the system allocates by default a single node worth of capacity as spare capacity. Storage data and the spare capacity are spread across all nodes in the protection domain. When a failure occurs, PowerFlex performs all the calculations to rebuild the data on a healthy device or node using the spare capacity.

One spare node is mandatory, but you can optionally deploy and configure two spare nodes to ensure sufficient capacity to rebuild two full nodes. In deployments with one spare node, if two nodes or devices fail, the second failure does not trigger a rebuild. Instead, the system continues to operate in a degraded state with some performance impact until the failed node is fixed.

The addition of one spare node translates into the following minimum storage node configuration, based on the protection scheme used:

- 2+2 protection scheme: a minimum of five nodes
- 4+2 protection scheme: a minimum of seven nodes
- 8+2 protection scheme: a minimum of 11 nodes

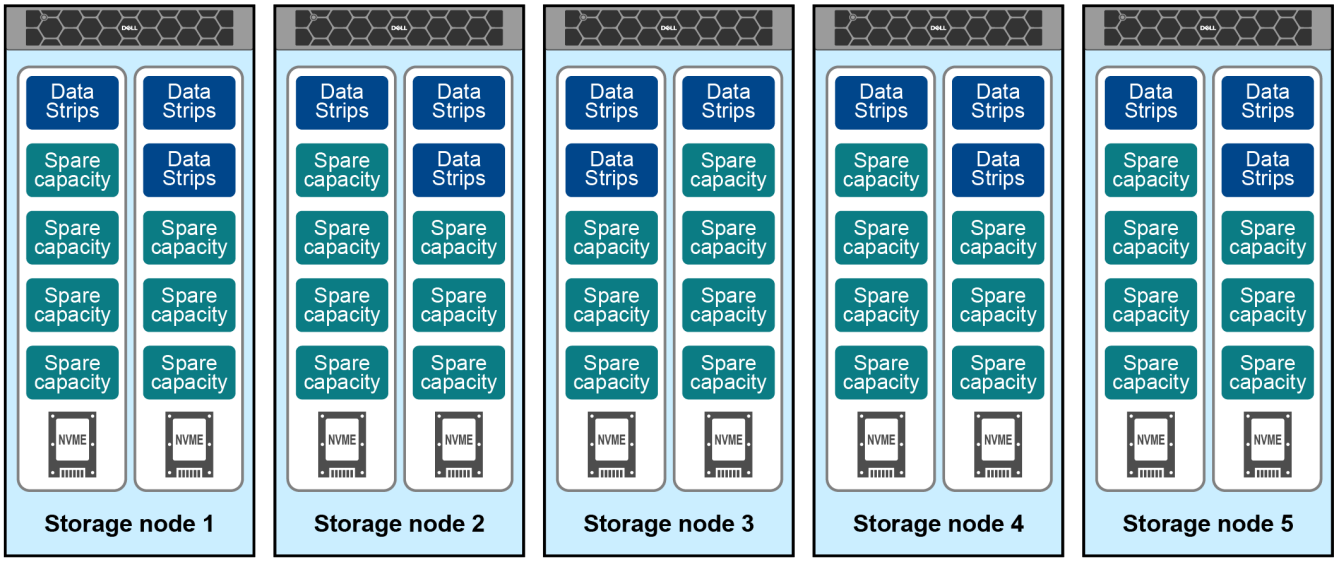


Figure 11. Example distribution of data strips and spare capacity in a 2+2 protection scheme

In an 8+2 protection scheme with 11 nodes, if each node is 10 TB, the spare capacity is 10 TB. Dell Technologies recommends that devices and nodes be uniform in size and that you avoid associating the device group with inconsistent devices sizes or a varied number of devices with each node. Inconsistent device sizes and numbers of devices per node may impact performance and some capacity may not be usable.

For example, if the largest device within a storage node is 900 GB and there are five devices in the node, spare capacity should be configured as 4.5 TB.

Rebuild

PowerFlex immediately initiates a process of protecting data when a storage node, device, or network fails. This data protection process is called a rebuild.

Rebuild is an automatic background process that recovers and rebuilds data from parity strips and writes the data to spare capacity on a healthy device. PowerFlex remains fully operational during a rebuild. User I/Os are not affected by the rebuild process, other than a possible slowdown in performance.

Rebuild uses reserved spare space to enable rebuild:

If there are two concurrent failures, PowerFlex uses the spare capacity and performs a rebuild based on a set of priorities. The first priority is the number of failed strips: A stripe with two failed strips is rebuilt before a stripe with one failed strip. This ensures optimal utilization of spare capacity at the time of rebuild and improves overall system availability.

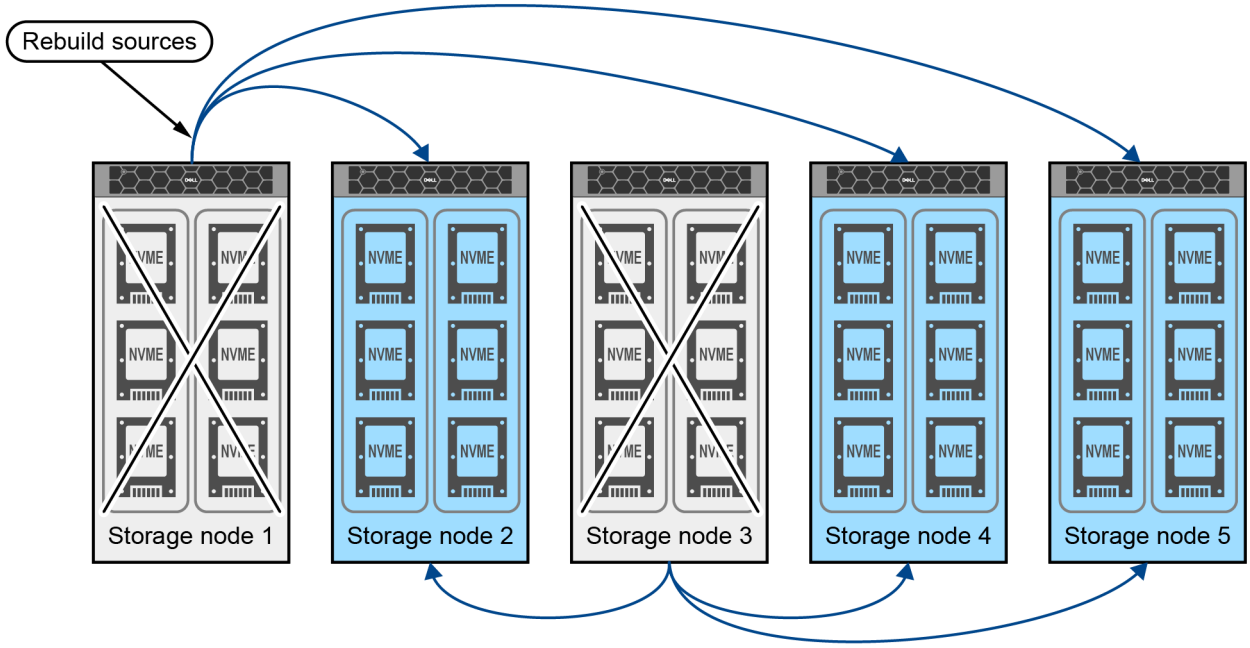


Figure 12. Rebuild process

PowerFlex also prioritizes rebuilding metadata over rebuilding data. Not only is the system impact of losing metadata greater than that of losing data, but metadata can be rebuilt more quickly than data.

If spare capacity is exhausted, rebuild does not occur unless a failed disk or set of disks is replaced with new ones.

Rebuild throttling

The rebuild throttling policy determines the priority of rebuild I/Os versus application I/Os when accessing storage devices. The possible rebuild throttling policies are no limit on rebuild I/Os, limit concurrent I/Os per storage device, favor application I/Os, and dynamic bandwidth throttling.

Rebuild throttling sets the rebuild priority policy for the storage pool. The policy determines the priority between the rebuild I/O and the application I/O when accessing storage devices. Application I/Os are continuously served.

Decreasing rebuild throttling increases the time that that it takes to rebuild the data from parity strips. But it also reduces the impact on the application. You must attempt to choose the right balance between the two.

The following priority policies can be applied:

- No limit: No limit on rebuilding I/Os. Any rebuild I/O is submitted to the device immediately, without further queuing.

NOTE: Rebuild I/Os are relatively large. Setting this policy speeds up the rebuild, but has the maximal effect on the application I/O.

- Favor application: Limit rebuild in both bandwidth and concurrent I/Os. The rebuild I/Os are limited both in bandwidth and in the number of concurrent I/Os. As long as the number of concurrent rebuild I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebuild I/Os are served. Once either threshold is reached, the rebuild I/Os wait until both I/O and bandwidth are below their thresholds. For example, setting the value to **1** guarantees that the device has one concurrent rebuild I/O at any given moment, which ensures that the application I/Os only wait for one rebuild I/O in the worst case. This imposes bandwidth on top of the limit concurrent I/Os option, which is a prerequisite to using this policy.
- Rebuild concurrent I/O limit default: 4 concurrent I/Os.

NOTE: Rebuild throttling affects system performance and should only be used by advanced users.

Rebalance

Rebalance is the process of relocating data to a different device or node. It occurs when PowerFlex detects that the user data is not evenly balanced across the nodes and devices.

The need for rebalance can occur as a result of several conditions, such as the addition or removal of a storage node, the addition or removal of a device, or recovery from a service, network, or device failure. PowerFlex moves copies of the data from the most used to the least used devices

The rebalance process does not run until the rebuild process is complete.

In the following image, the data is spread across four nodes in the protection domain before rebalance. Following rebalance, the storage data is balanced across five nodes. For the sake of simplicity, the image displays only a small amount of the data on each node. In actuality, data and spare storage are allocated evenly following rebalance.

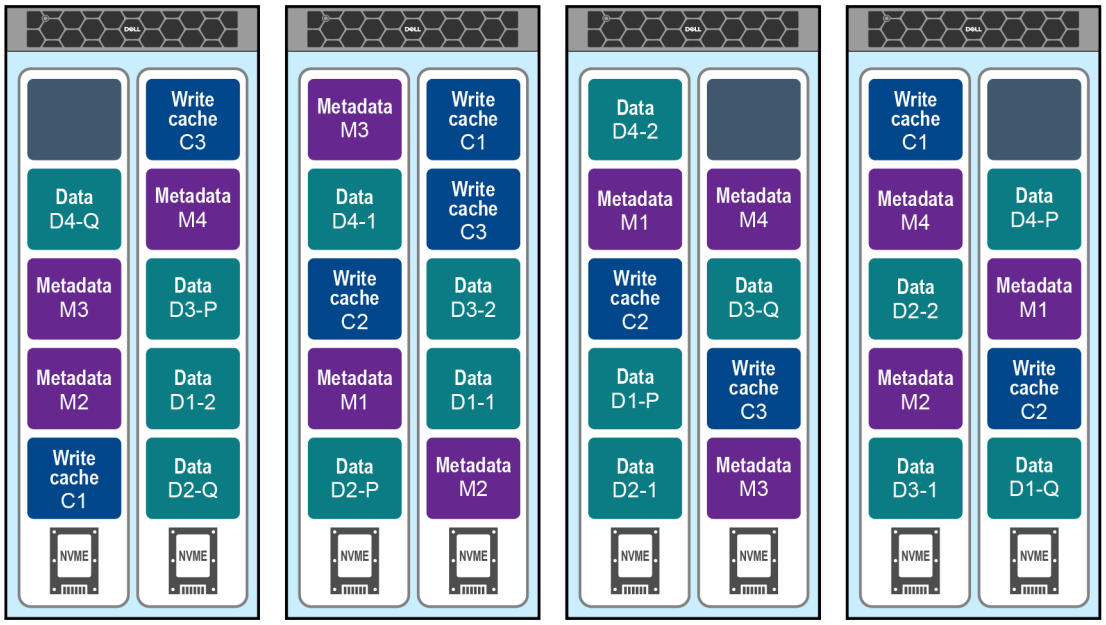


Figure 13. Rebalance process: Storage before rebalance

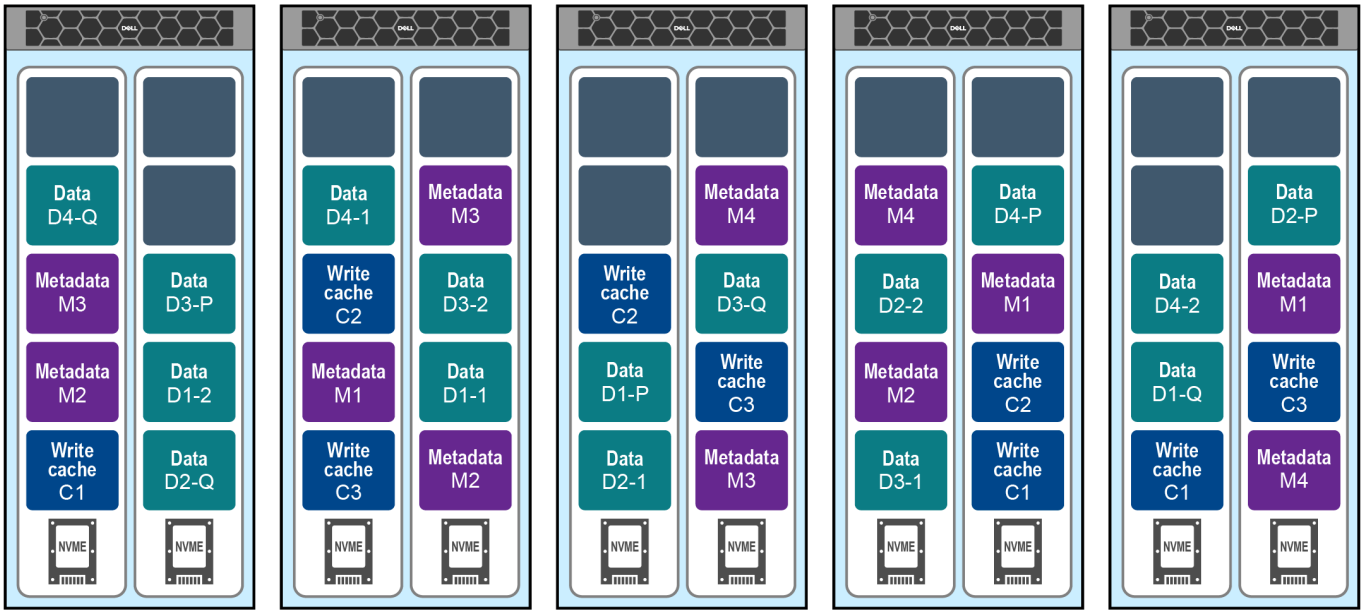


Figure 14. Rebalance process: Storage after rebalance

Rebalance throttling

The rebalance throttling policy determines the priority of rebalance I/Os versus application I/Os when accessing storage devices.

Rebalance throttling sets the rebalance priority policy for the storage pool. The policy determines the priority between the rebalance I/O and the application I/O when accessing storage devices. Application I/Os are continuously served. Rebalance, unlike rebuild, does not impact the reliability of the system, and reducing its impact is not risky.

NOTE: Rebalance throttling affects the performance of the system, and should be used only by advanced users.

The following possible priority policies can be applied:

- No limit: No limit on rebalance I/Os. Any rebalance I/O is submitted to the device immediately, without further queuing.

  NOTE: Rebalance I/Os are relatively large. Setting this policy speeds up the rebalance, but has the maximal effect on the application I/O.

- Favor application: Limit rebalance in both bandwidth and concurrent I/Os. The rebalance I/Os are limited both in bandwidth and in the number of concurrent I/Os. As long as the number of concurrent rebalance I/Os, and the bandwidth they consume, do not exceed the predefined limits, rebalance I/Os are served. Once either limit is reached, the rebalance I/Os wait until the limits are no longer met. This imposes a bandwidth limit on top of the limit concurrent I/Os option.
- Default values:
  - The default policy for rebalance: Favor application.
  - Rebalance concurrent I/O limit: 4 concurrent I/Os per storage device.
  - Rebalance bandwidth limit: 40960 KB/s.

Checksum protection

PowerFlex calculates and validates the checksum value for the payload during transit to protect data in flight. Checksum protection is applied to all I/Os (application, rebuild, and rebalance). The feature is enabled by default and cannot be disabled.

6

PowerFlex management platform

The PowerFlex management platform provides the management and orchestration in a single tool for all PowerFlex functions, from deployment through administration and life cycle management.

The PowerFlex management platform is a management software stack that is based on the Rancher Kubernetes Engine (RKE). It is designed as microservices, with specialized modules working together. Installed across three nodes, the PowerFlex management platform provides high availability, resiliency, and reliability.

A web user interface, PowerFlex Manager, provides ease of access to the PowerFlex management platform. You can also access PowerFlex management platform functionality through the PowerFlex REST API or the PowerFlex CLI (SCLI).

Several PowerFlex management platform deployment configurations are supported, depending on the PowerFlex consumption model. These configurations include installing the PowerFlex management platform on the same server as a PowerFlex storage node in an integrated PowerFlex management (non-KVM) deployment configuration, and the PowerFlex management cluster for PowerFlex appliance and PowerFlex rack.

Capabilities

The PowerFlex management platform includes the following features and capabilities:

Interfaces

- PowerFlex Manager: A single web user interface for full lifecycle mode services
- PowerFlex REST API: A single REST endpoint for all APIs, including legacy and the modern ISG-standard Power API
- PowerFlex CLI (SCLI)

User management

- Role-based and scope-based access control and user management, with a single set of roles.
- Supports both PowerFlex local users and LDAP and Active Directory users, without the need for additional configuration.
- Supports common PowerAPIs to improve the ecosystem and customer integration of the PowerFlex storage solution.

PowerFlex Manager unified services

While PowerFlex can be managed through the REST API or SCLI, PowerFlex Manager supports a full range of services:

- Security
- Single sign-on (SSO) and role-based access control
- Deployment and lifecycle management
- Events, alerts, and reporting
- Events (log) and alerts (state), including meta data manager (MDM) events
- Logging and metrics collection
- SNMP, secure connect gateway, SMTP, Dell AIOps

Security

The PowerFlex management platform supports system security, as follows:

- The PowerFlex management platform exposes an HTTPS interface for web user interfaces and REST APIs.
- Users must have an account that is defined in PowerFlex.

- Upon login, users receive an access token.
- PowerFlex Manager uses OpenID Connect (OIDC).
- Communications between the PowerFlex management platform and the PowerFlex block storage system are secured using mTLS.
- The strength of security is enhanced by using SSH keys and non-root users through sudo in the management aspects of PowerFlex.

CLI tools

The following CLI tools are available:

- A CLI that provides full control over the block storage system.
- A CLI with expanded support for management, reporting, and alerts.

Alerts

PowerFlex supports webhooks, enabling the system to send alerts to webhook servers such as BigPanda.

Additional features

- PowerFlex management platform runtime is optimized to 60 minutes.

PowerFlex appliance and PowerFlex rack

PowerFlex appliance and PowerFlex rack include the following additional features for the PowerFlex management platform:

- Multi-subnet and VLAN support allows defining multiple subnets and VLANs in a single network. The most common implementation example is with a management or vMotion network that spans multiple racks. Each rack is its own subnet and VLAN.

  NOTE:

  Multi-subnets and VLANs are not supported on PowerFlex data, vSAN, and NSX external networks.

  Multi-subnets and VLANs are not supported in deployments with the PowerFlex management cluster (KVM).

- The PowerFlex management cluster on PowerFlex appliance and PowerFlex rack enables autonomous monitoring of the underlying controller system.
- The PowerFlex management cluster supports end-to-end upgrades, with alerts regarding issues encountered in the PowerFlex management cluster infrastructure.
- Modular, single-component upgrades change the package in the PowerFlex rack and PowerFlex appliance catalogs to newer or older versions.

  This upgrade feature is for cases where customers want to deviate from the Release Certification Matrix (RCM) or Intelligent Catalog (IC), such as applying security patches without performing a full RCM or IC upgrade.

  NOTE: By default this feature is turned off and can only be enabled through an RPQ process. Contact your Dell Technologies account representative for details.

PowerFlex Manager

PowerFlex Manager is the unified management user interface for all PowerFlex systems. It allows you to deploy and manage new and existing PowerFlex systems, and also includes lifecycle and resource management capabilities.

PowerFlex Manager provides IT operations management for PowerFlex systems. It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. PowerFlex Manager can also be used to deploy and manage new and existing PowerFlex appliance and PowerFlex rack systems.

PowerFlex Manager includes the following tabs:

- A dashboard that enables you to see a summary of performance and PowerFlex appliance and PowerFlex rack inventory data.
- A block tab that provides a user interface for managing PowerFlex block storage.
- A protection tab that provides a user interface for performing PowerFlex storage protection operations.
- A lifecycle tab that allows you to create and clone templates for PowerFlex appliance and PowerFlex rack and deploy them as resource groups (previously known as services).
- A resources tab that allows you to manage all resource types within the system.
- A monitoring tab that allows you to track events, alerts, and jobs that are running.
- A settings tab that enables you to configure system settings for PowerFlex Manager.

PowerFlex Manager is part of all PowerFlex consumption models.

Serviceability

PowerFlex Manager provides automated and remote support through integration with Dell Technologies connectivity services, which ensures better alignment with Dell Technologies services initiatives to enhance the user experience.

Registering your device with Dell Technologies connectivity services allows Dell Technologies connectivity services to:

- Collect system state information and telemetry.
- Automate issue detection and support case creation.
- Resolve common issues remotely.
- Provide critical information to Dell Technologies remote support to resolve complex issues.

Deployment options for each PowerFlex consumption model

The PowerFlex management platform is an integral part of PowerFlex. The PowerFlex management platform can be set up in multiple deployment options for on-premises deployments, regardless of the consumption model.

Depending on which PowerFlex consumption model you choose, you have different management options within the solution stack. PowerFlex Manager manages some aspects of the solution, while other elements are customer managed.

NOTE: PowerFlex for AWS and PowerFlex for Microsoft Azure require dedicated AWS EC2 instances or Azure virtual machines, respectively. Deployment details are explained in the *PowerFlex for AWS Deployment Guide* and *PowerFlex for Microsoft Azure Deployment Guide* for the specific offering.

The following table describes the supported deployment options for the PowerFlex management platform:

Table 17. PowerFlex management platform deployment options

Deployment offering	PowerFlex software	PowerFlex appliance	PowerFlex rack	PowerFlex custom node	Description
PowerFlex management cluster (KVM)	Not supported	Supported	Supported	Not supported	The management cluster runs a dedicated instance of PowerFlex Manager to manage it with a minimum of three nodes. Both the PowerFlex and PowerFlex Manager versions running on the management cluster are 5.x. A second instance of PowerFlex Manager manages the PowerFlex storage clusters.
Single ESXi PowerFlex management node	Not supported	Supported	Not supported	Supported	VMware ESXi-based single node with local RAID 5 storage solution.
Customer-provided infrastructure	Supported	Supported	Not supported	Supported	Allows you to supply a hypervisor to host the management virtual machines.

Table 17. PowerFlex management platform deployment options (continued)

Deployment offering	PowerFlex software	PowerFlex appliance	PowerFlex rack	PowerFlex custom node	Description
for PowerFlex management (ESXi)					
Customer-provided infrastructure for PowerFlex management (KVM/Linux)	Supported	Supported	Not supported	Supported	Allows you to supply a hypervisor to host the management virtual machines.
Customer-provided infrastructure for PowerFlex management (KVM/Linux) This offering requires special approval. It is not a standard offering.	Supported	Supported	Not supported	Supported	Allows you to supply three bare-metal Linux-based servers to host the PowerFlex management platform.
Integrated PowerFlex management (non-KVM)	Supported	Not supported	Not supported	Supported	The PowerFlex storage nodes host the management instances of Kubernetes clusters and containers in a co-residency configuration.

NOTE: PowerFlex for AWS and PowerFlex for Microsoft Azure require dedicated AWS EC2 instances or Azure virtual machines, respectively. Deployment details are explained in the *PowerFlex for AWS Deployment Guide* and *PowerFlex for Microsoft Azure Deployment Guide* for the specific offering.

For deployment details for the integrated PowerFlex management (non-KVM) cluster for PowerFlex on-premises deployment offerings for PowerFlex software and PowerFlex custom node, see the *PowerFlex and PowerFlex Manager Software Install Guide*.

Dedicated management cluster model

The PowerFlex management platform can be deployed on dedicated management nodes or clusters running outside of PowerFlex storage nodes and PowerFlex compute nodes. Both the PowerFlex and PowerFlex Manager versions running on the management cluster are 5.x.

When you deploy a dedicated management cluster, the PowerFlex management platform manages the complete solution stack. In the following diagram, the dedicated management nodes are in clusters running outside of PowerFlex compute-only nodes and PowerFlex storage-only nodes.

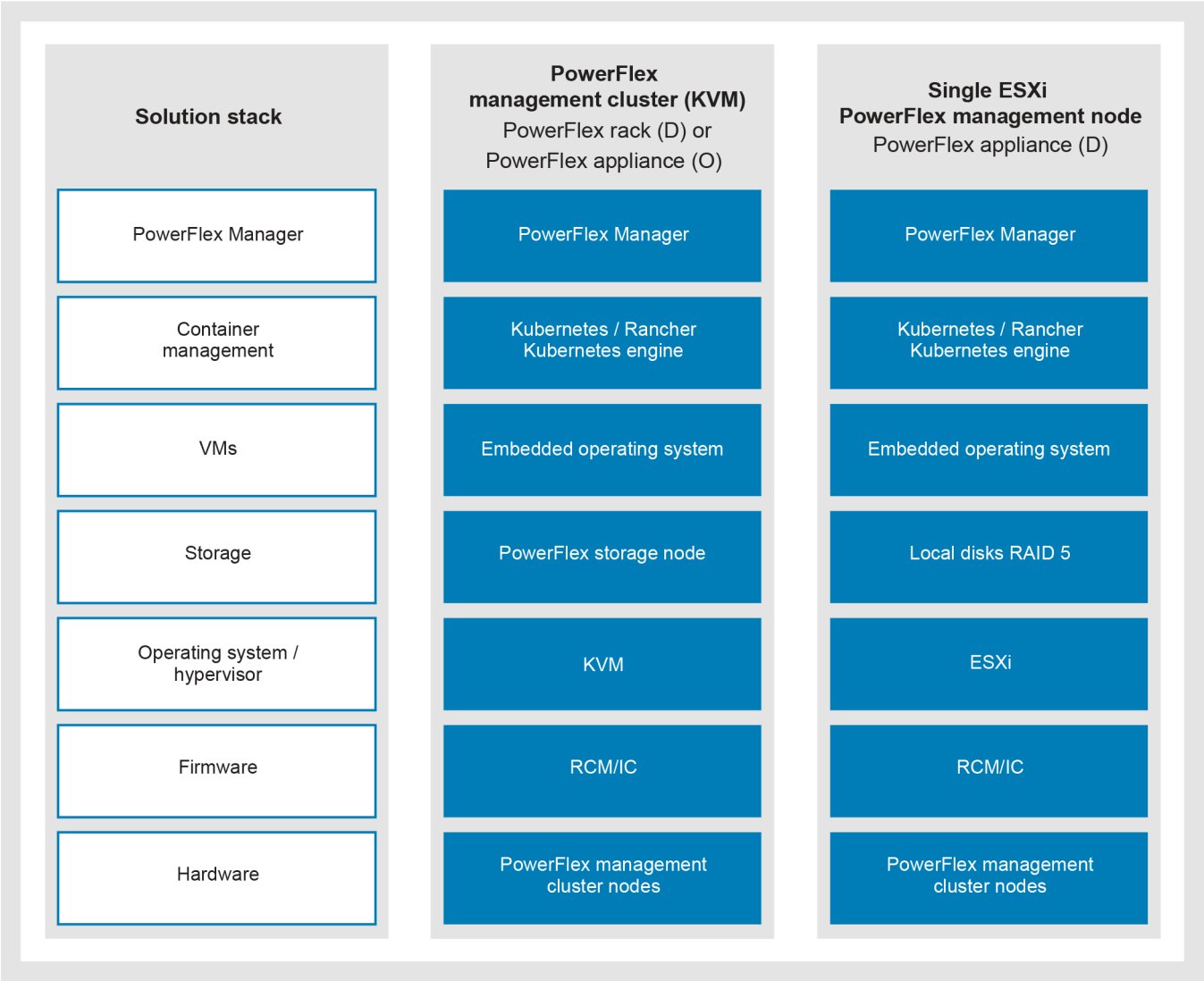


Figure 15. Dedicated management cluster model

Legend

PowerFlex Manager manages these components.

(D): default configuration

(O): optional configuration

Integrated PowerFlex management (non-KVM)

The PowerFlex management platform supports an integrated PowerFlex management (non-KVM) deployment model in which the PowerFlex management platform is deployed on the same machines as PowerFlex storage-only nodes for PowerFlex software and PowerFlex for Public Cloud.

In the integrated PowerFlex management (non-KVM) deployment model, the PowerFlex management platform is deployed on the boot device of three PowerFlex storage nodes so that the PowerFlex management platform and PowerFlex storage co-reside on the same node. For more information, see the *PowerFlex and PowerFlex Manager Software Install Guide*.

In this management option, the PowerFlex management platform manages only the containers and PowerFlex, as illustrated in the following diagram:

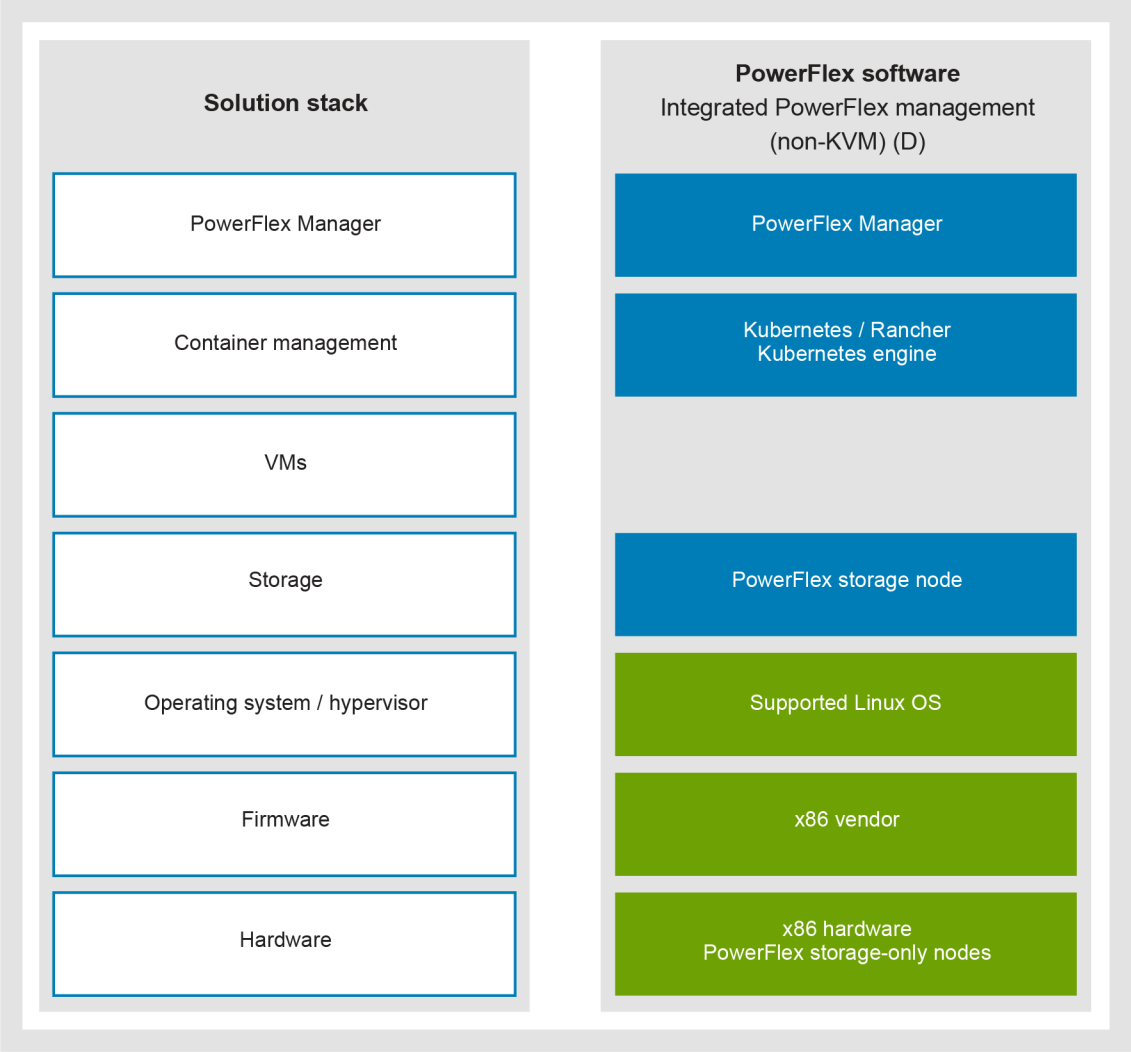


Figure 16. Integrated PowerFlex management (non-KVM) model

Legend

PowerFlex Manager manages these components.

Users manage these components.

(D): default configuration

Customer-provided infrastructure for PowerFlex management

The PowerFlex management platform can be deployed on a supported operating system of your choice. In this management option, the PowerFlex management platform manages only the containers and PowerFlex.

The following image explains the solution stack, depending on the operating system:

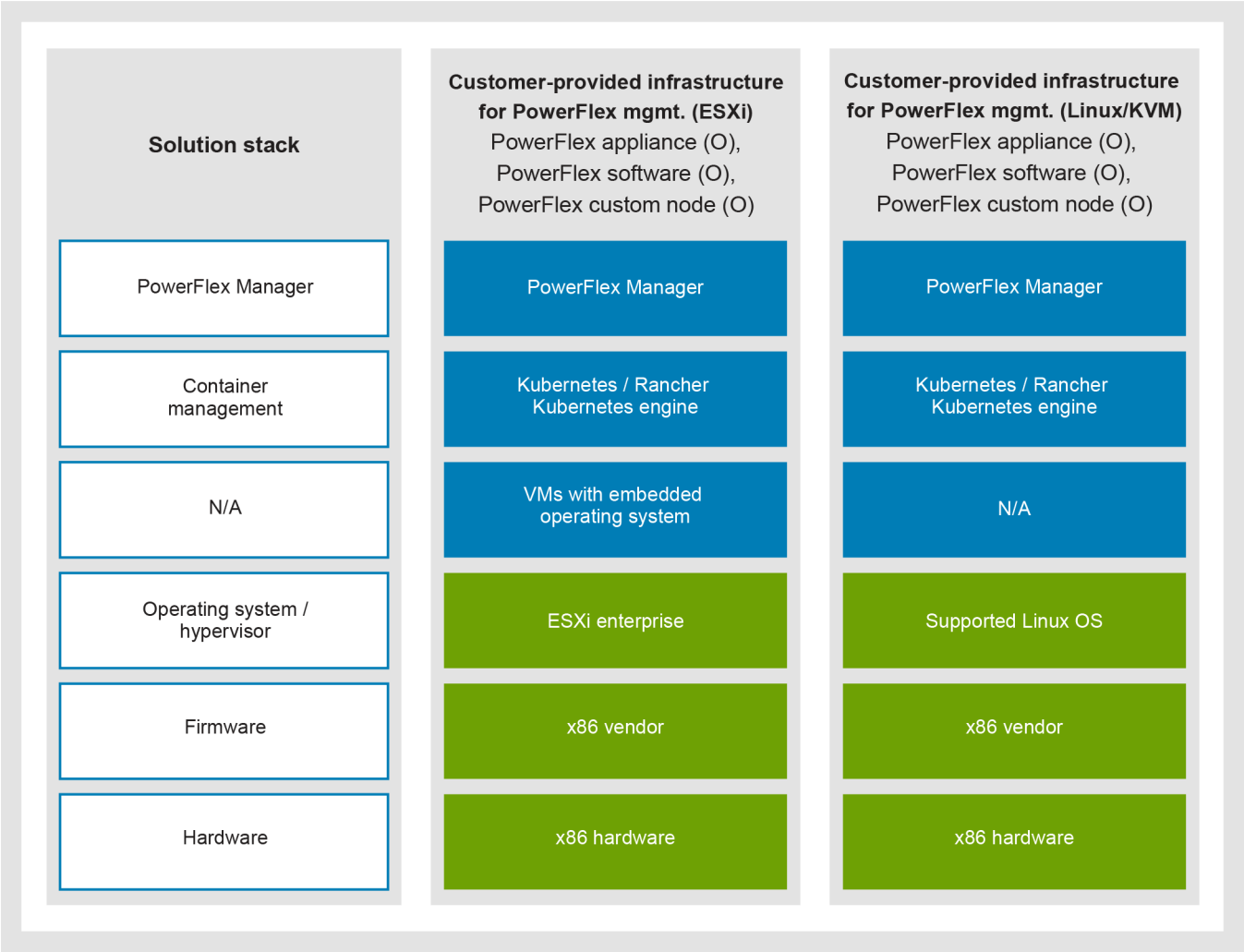


Figure 17. Customer-provided infrastructure for PowerFlex management model

Legend

PowerFlex Manager manages these components.

Users manage these components.

(O): optional configuration

7

Networking

In PowerFlex, inter-node communications manage data locations, rebuild, rebalance, and access of applications to stored data on one IP network, or on separate IP networks.

Use any of the management interfaces to perform management tasks in one of the following ways:

- Through a separate network, with access to the other PowerFlex components.
- On the same network.

You can configure these options during deployment using the CSV topology file (or a PowerFlex Manager template for PowerFlex appliance or PowerFlex rack, or after deployment using the SCLI (PowerFlex CLI).

Network requirements

The following table includes the list of PowerFlex network requirements:

Table 18. Network requirements

Item	Requirement
Network type	Ethernet based
Minimum supported speed	2*25 GbE
Recommended speed	4*25 GbE
Certified configurations	4*25 GbE, 2*100 GbE, 4*100 GbE
Supported MTU	1500/9000 (jumbo frames recommended for optimal performance)
High availability and load-balancing	At least two physical ports with: ● PowerFlex native multipath across two or more subnets. (Up to eight subnets supported.) ● Link aggregation. A single IP subnet can be used with link aggregation; otherwise a minimum of two subnets are required. Dell Technologies recommends that you use MDM virtual IPs for simplified network stability in the event of failures.
IP addresses	For more information about calculating IP addresses, see *PowerFlex Appliance Network Planning Guide*.
Data network latency requirements for MDMs	Maximum of 300 milliseconds
Switch redundancy	● Switches must be redundant to provide continued access to components inside the rack in the network in case an access switch fails. ● Flat network is supported. ● Leaf-spine architecture is recommended for cross-rack traffic. ● VXLAN is supported.
SDC connectivity	Connection through a router if supported and required for a low-latency connection for optimal performance.
PowerFlex single management subnet	Dedicated port or a VLAN on a trunk connection. For best practices, see the *PowerFlex Apppliance Network Planning Guide*.

Network limitations

The following table describes the PowerFlex 5.x network limitations:

Table 19. Network limitations

Item	Limitation
RMDA	Not supported
InfiniBand	Not supported
IPv4	Supported on all PowerFlex offerings.
IPv6	Supported only on PowerFlex software offerings.
DHCP	Cannot be deployed on a network in which PowerFlex MDMs, storage nodes, storage data targets, or storage data clients reside.
Firewall rules	Must allow PowerFlex ports. For port information, see the *PowerFlex Family 5.x Security Configuration Guide*.

Network considerations

The following table explains the PowerFlex network configurations and the benefits of each of them:

Table 20. Network configurations

Network type	Benefits
Single IP network All communications and I/Os used for management and for data storage are performed on the same IP network.	• Ease of use • Fewer IP addresses required
Multiple separate IP networks Separate networks are used for management and for data storage, or separate networks are used within the data storage part of the system.	• Security • Redundancy • Performance • Separate IP roles to enable separation between customer data and internal management

NOTE: Network high availability can be implemented by using NIC-bonding.

IP address configurations

The following table describes the different IP addresses for PowerFlex, and the IP address configurations:

NOTE: The storage node contains the PowerFlex data server (PDS) and device gateway target (DGWT) components.

Table 21. IP addresses for PowerFlex

Address type	Purpose	Description
MDM management IP address	Management access	• Optional, but recommended. • Applicable to MDM and storage nodes that can be used to provide access to PowerFlex management applications, such as SCLI and the REST API. • This IP address must be externally accessible.

Table 21. IP addresses for PowerFlex (continued)

Address type	Purpose	Description
MDM IP addresses	Control network	• Mandatory IP addresses used for MDM control communications with storage nodes and SDCs. • Used to convey data migration decisions, but no user data passes through the MDM. • Must be on the same network as the data network. • MDM cluster IP address limit is 16 IP addresses, which include all cluster members (primary, secondary, and standby).
Storage node IP role - all PowerFlex products	Rebuild/rebalance and data path network	• IP addresses used for both storage node-to-storage node and storage node-to-SDC communications. • These IP addresses are also used to communicate with the MDM.
Storage node IP role - storage only	Rebuild/rebalance network	• IP addresses used for storage node-to-storage node communications only. Storage node-to-storage node communications are communications between : ○ PowerFlex data server (PDS) and device gateway target (DGWT) ○ PDS and MDM ○ DGWT and MDM • These addresses are used for rebuild and rebalance operations. • These IP addresses are also used to communicate with the MDM.
Storage node IP role - application only	Data path network	• IP addresses used for storage node-to-SDC communications. Storage node-to-SDC communications are communications between the SDC and PDS. • These addresses are only used for read/write user data operations.

Network limitations when using VMware ESXi

When deploying PowerFlex on VMware ESXi, configurations with more than one vmknic interface on the same IP subnet are not supported.

For more information, see vSphere Networking.

PowerFlex only supports the following network configurations when deployed on ESXi:

- A single data storage network - supported in only the PowerFlex software and PowerFlex custom node product offerings.
- Two or more data networks, each on separate IP subnets
- One or more data storage networks using several NIC-bonding configurations or load-balancing methods.

Network stability

Resilient network connectivity is standard for PowerFlex. Nevertheless the actual implementation may vary. Sometimes, resiliency is attained through LACP or bonding, while in other cases through application-level multipathing.

PowerFlex has additional capabilities to support partial failures on existing connections. While many network issues are manifested as a complete disconnection, some are not. PowerFlex is aware of degraded connections, including:

- Stable, but slow connections in relation to other connections.
- Unstable connections (occasional disconnections).
- A combination of the above two.

PowerFlex supports using the MDM secondary IP addresses as a proxy connection to the primary MDM.

NOTE: This capability is applicable only when the MDM VIP is not in use, because it requires the SDCs to have all the IP addresses configured to attempt a proxy connection.

8

Implementing PowerFlex

PowerFlex is based on a configuration of hardware and software components working together.

The PowerFlex physical layer consists of the hardware (servers with storage devices and the network between them) and the PowerFlex software that is installed on them. Communications are carried out over the existing LAN data network using standard TCP over IP. The MDM and storage nodes can be assigned to up to eight IP addresses, enabling wider bandwidth and better I/O performance and redundancy.

Implementing PowerFlex is, in general, a two-step process: first build the physical storage layer, then configure the virtual SAN layer on top of it.

Physical layer

To implement the physical layer, perform the following steps:

1. Physically assemble the server nodes in the rack and perform all cabling, according to the instructions for your PowerFlex product.
2. Configure the operating system, including all required IP addresses, based on network best practice guidelines.
3. Deploy the PowerFlex software.

The physical layer is now ready, and you can expose a virtual storage layer.

1. Install the MDM component on the MDM nodes in a five-node redundant cluster configuration (one primary MDM, two secondary MDMs, and two tiebreakers). MDMs do not require dedicated nodes. They can be installed on nodes hosting other PowerFlex components.

   NOTE: ActiveMQ is installed only on the primary and secondary MDMs.

2. Install the lightweight installation agent (LIA). The LIA is a required component for many maintenance operations.
3. Install the storage node components (PowerFlex data server and device gateway target) on all nodes that contribute some, or all, of their physical storage.
   - Divide the storage nodes into protection domains. Each storage node can be a member of only one protection domain.
   - Divide the physical storage units in each protection domain into device groups and storage pools.

   NOTE: PowerFlex supports only one storage device group and one persistent memory device group per protection domain in this release.

4. Install the SDC component or configure the NVMe over TCP client on all nodes on which the application accesses the data exposed by the PowerFlex volumes.

The following figure shows a PowerFlex compute-only and storage-only components.

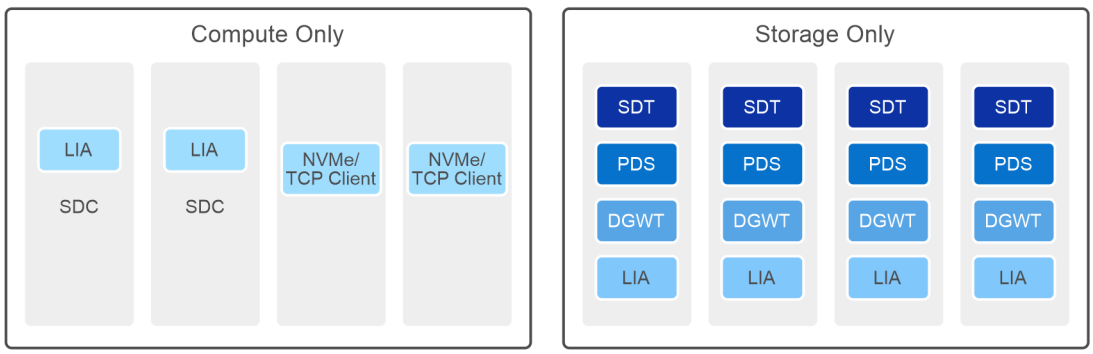


Figure 18. PowerFlex cluster in a mixed configuration

NOTE: PowerFlex appliance and PowerFlex rack offer more automated installation experiences using PowerFlex Manager. For more information, see PowerFlex appliance and PowerFlex rack product information.

Deploying MDMs in a multirack system

The following list includes best practices for adding MDMs to a multiple-rack system:

- For a large system with multiple racks, the MDM cluster members should be distributed in separate racks for resiliency.
- The MDM network requirements described in the Networking section must also apply to the network connections between the racks.
- To ensure that users do not shut down the MDMs by mistake, mark them physically with stickers saying **Do not shutdown: MDM installed here**, or add a meaningful prefix or suffix to the hostname of the server where MDM is installed. For example: "Rack22U32-MDM." You can also confirm that a different node in the rack is used for each server, as shown in the following table:

Table 22. Nodes in rack

Node level	Rack number	MDM role
Top node	1	MDM
Bottom node	2	MDM
Middle node	3	TB

Approved RAID controller

The following is the RAID controller that is tested and certified for this version of PowerFlex.

PowerFlex-certified RAID controller

- Manufacturer: Dell
- Specifications:
  - Model name: PowerEdge RAID Controller H755
  - Firmware version: see PowerFlex custom node Firmware and Driver support matrix.
  - Driver version: see PowerFlex custom node Firmware and Driver support matrix.
  - Driver name: megaraid_sas

SAN virtualization layer

After you install and configure the storage data client (SDC) components or NVMe hosts, expose the virtual SAN devices to the servers on which the SDCs or NVMe over TCP clients are installed.

The MDM cluster manages the entire PowerFlex system. It aggregates the entire storage that is exposed to it by all the storage nodes to generate a virtual layer, which is the virtual SAN storage. Volumes can now be defined over the storage pools and exposed to the applications as a local storage device using the SDCs or NVMe hosts.

To expose the virtual SAN devices to the servers on which the SDCs and NVMe hosts are installed, perform the following:

1. Define volumes. Each volume that is defined to a storage pool is evenly distributed over all members using a RAID protection scheme. By having all storage node members of the storage pool participate, PowerFlex ensures:
   - The highest, most stable, and consistent performance possible.
   - Rapid recovery and redistribution of data.
   - Massive IOPS and throughput.

   All the volumes that are allocated in PowerFlex are thin volumes.
2. Map the volumes. Designate which SDCs or NVMe hosts can access the given volumes so that:
   - Access control per volume is exposed.
   - All or no volumes are shared.

   After an SDC or NVMe host is mapped to a volume, it immediately gets access to the volume and exposes it locally to the applications as a standard block device. These block devices appear as `/dev/sciniX` where *X* is a letter, starting from `a`.

For example:

`/dev/scinia`

`/dev/scinib`

Each `/dev/sciniX` volume can have a maximum of 15 partitions.

For the maximum number of volumes that can be mapped to an SDC or NVMe host, see the *Supported capabilities and system limits* table in related information.

NOTE: SDC mapping is similar to LUN mapping, in the sense that it only allows volume access to clients that were explicitly mapped to the volume.

Related information

Product limits

Audit logging

PowerFlex enables the redirection of the PowerFlex events and Ingress audit messages to syslog.

These events and audit messages are forwarded to an external Security Information and Event Manager (SIEM). You must perform the following configuration steps to enable this feature in PowerFlex Manager:

- Change Ingress settings to emit audit messages.
- Define a notification policy to forward Ingress audit messages in the PowerFlex system to an SIEM.
- Define a notification policy to forward events in the PowerFlex system to an SIEM.

For more information about the configuration steps, see the *PowerFlex and PowerFlex Manager Software Administration Guide*.

Supported events

Audit logging supports the following event categories that occur in PowerFlex Manager and integrated components and systems:

- User authentication and authorization
- Changes to system configuration and security policies
- Security events

Implementation considerations

The following details should be taken into account when implementing PowerFlex:

Configuring volumes

You must add and map volumes for applications to be able to access them. All volumes are thin volumes.

Define volume names according to the following rules:

- Contains fewer than 32 characters.
- Contains only alphanumeric and punctuation characters.
- Is unique within the object type.

Dell Technologies recommends that you give each volume a meaningful name associated with its operational role.

PowerFlex objects are assigned a unique ID that can be used to identify the object in CLI commands. You can retrieve the ID using a query, or using the property sheet for the object in PowerFlex Manager.

Naming PowerFlex objects

Dell Technologies recommends that you assign meaningful, user-defined names to all PowerFlex objects. Using meaningful names make it easier to perform actions such as defining volumes and associating them with applications.

For example, storage pools can be named *Capacity_Storage* and *Performance_Storage*, which allows you to identify the different tiers.

For protection domains, one example would be to separate the storage nodes used by a finance department from storage nodes used by the engineering department by naming the protection domains *Financial-PD* and *Engineering-PD*. This separation of different departments provides many benefits, including security.

Related information

Volumes

Consumption models

PowerFlex is offered in the following consumption models:

- PowerFlex software
- PowerFlex appliance
- PowerFlex rack
- PowerFlex custom node
- PowerFlex for AWS
- PowerFlex for Microsoft Azure

PowerFlex software

The following table describes the components in the PowerFlex software offering:

Table 23. PowerFlex software components

Component	Software management catalog
PowerFlex software	Included
Operating system	Not included. See the *PowerFlex Support Matrix*.
System firmware • BIOS • Cards • Drives	Customer managed
Drivers	Customer managed
Infrastructure firmware • Cabinet/PDU • IPI Appliance	N/A
Dell CloudLink	See the *PowerFlex Support Matrix*
VMware NSX	Customer managed
Network switch firmware and operating system	Customer managed

PowerFlex appliance

PowerFlex appliance is a scalable system with flexible form factors that comes preconfigured and validated for fast, easy deployment.

The following table describes PowerFlex appliance components:

Table 24. PowerFlex appliance components

Component	PowerFlex appliance Intelligent Catalog
PowerFlex software	Included
Operating system	• VMware ESXi - not included • Embedded operating system - included
Drivers	Included
System firmware • BIOS • Cards • Drives	Included
Infrastructure firmware • Cabinet/PDU • IPI appliance	N/A
Dell CloudLink	Included
VMware NSX	Customer managed
Network switch firmware and operating system	• Full network automation (FNA) model - firmware included • Partial network automation (PNA) model For a complete list of supported hardware, go to the Code & Compatibility Knowledge Hub.

NOTE: The VMware ESXi ISO and the upgrade files are available for download from the Broadcom support portal.

PowerFlex rack

PowerFlex rack is a rack-scale system that is manufactured, managed, supported, and sustained as one system for single-end-to-end lifecycle support.

The following table describes the PowerFlex rack components:

Table 25. PowerFlex rack components

Component	PowerFlex rack Release Certification Matrix
PowerFlex software	Included
Operating system	• VMware ESXi - not included • Embedded operating system - included
Drivers	Included
System firmware • BIOS • Cards • Drives	Included
Infrastructure firmware • Cabinet/PDU • IPI Appliance	Included
Dell CloudLink	Included
VMware NSX	Customer managed
Network switch firmware and operating system	Included

NOTE: The VMware ESXi ISO and the upgrade files are available for download from the Broadcom support portal.

PowerFlex custom node

Table 26. PowerFlex custom node components

Component	Software management catalog
PowerFlex software	Included
Operating system	Not included. For a complete list of supported hardware, go to the Code & Compatibility Knowledge Hub.
System firmware • BIOS • Cards • Drives	Included in a separate OpenManage Enterprise catalog and driver bundle. For a complete list of supported hardware, go to the Code & Compatibility Knowledge Hub.
Drivers	Customer managed
Infrastructure firmware • Cabinet/PDU • IPI Appliance	N/A
Dell CloudLink	Not included. For a complete list of supported hardware, go to the Code & Compatibility Knowledge Hub.
VMware NSX	Customer managed
Network switch firmware	Customer managed

Dell PowerFlex for Public Cloud

The following table describes the components in the PowerFlex for Public Cloud offerings:

Table 27. PowerFlex for Public Cloud software components

Component	Details
PowerFlex software	Included
Operating system	PowerFlex for AWS: Available in AWS Marketplace. PowerFlex for Microsoft Azure: Available in Azure Marketplace
Dell CloudLink	PowerFlex for AWS: Available in AWS Marketplace. PowerFlex for Microsoft Azure: Available in Azure Marketplace

PowerFlex implementation in a VMware ESXi-based system

In a VMware ESXi environment, the storage data client (SDC) or NVMe host resides directly on the VMware ESXi host.

Supported offerings

For a full list of supported hypervisors and operating systems, see the Dell PowerFlex Simple Support Matrix . The following table describes the possible VMware ESXi-based configuration types for PowerFlex and the supported PowerFlex offerings for each configuration.

Table 28. Supported offerings for VMware ESXi-based system

Configuration type	Supported VMware ESXi versions	Supported PowerFlex offering
Managed VMware ESXi compute nodes with SDC connecting to a PowerFlex storage-only system	VMware ESXi 8.0 Dell Technologies recommends implementing the latest update.	PowerFlex appliance PowerFlex rack
Unmanaged VMware ESXi compute nodes with SDC connecting to a PowerFlex storage-only system	VMware ESXi 8.0 and 9.0 Dell Technologies recommends implementing the latest update.	PowerFlex software PowerFlex custom node PowerFlex appliance PowerFlex rack
VMware ESXi compute nodes with NVMe over TCP connecting to a PowerFlex storage-only system	VMware ESXi 8.0 and 9.0 Dell Technologies recommends implementing the latest update.	PowerFlex software PowerFlex custom node PowerFlex appliance PowerFlex rack

Components in a PowerFlex VMware ESXi-based implementation

The PowerFlex compute-only configuration on a VMware ESXi host consists of the following software components:

- Storage data client (SDC): A lightweight device driver that presents PowerFlex volumes as block devices to the application on the server on which the SDC is installed. The SDC creates a logical adapter, which is a VMware ESXi kernel construct. The adapter informs VMware ESXi about the arrival and disappearance of SCSI devices. These LUNs can be formatted with VMFS and then exposed using the VMware ESXi host to the virtual machine or can be used as RDM devices.
- Lightweight installation agent (LIA): A required component for many maintenance operations. During initial installation, the installation manager establishes a trust with the LIA, which facilitates its operation.

In the following diagram, the SDC is installed within the VMware ESXi kernel similar to any other VIB.

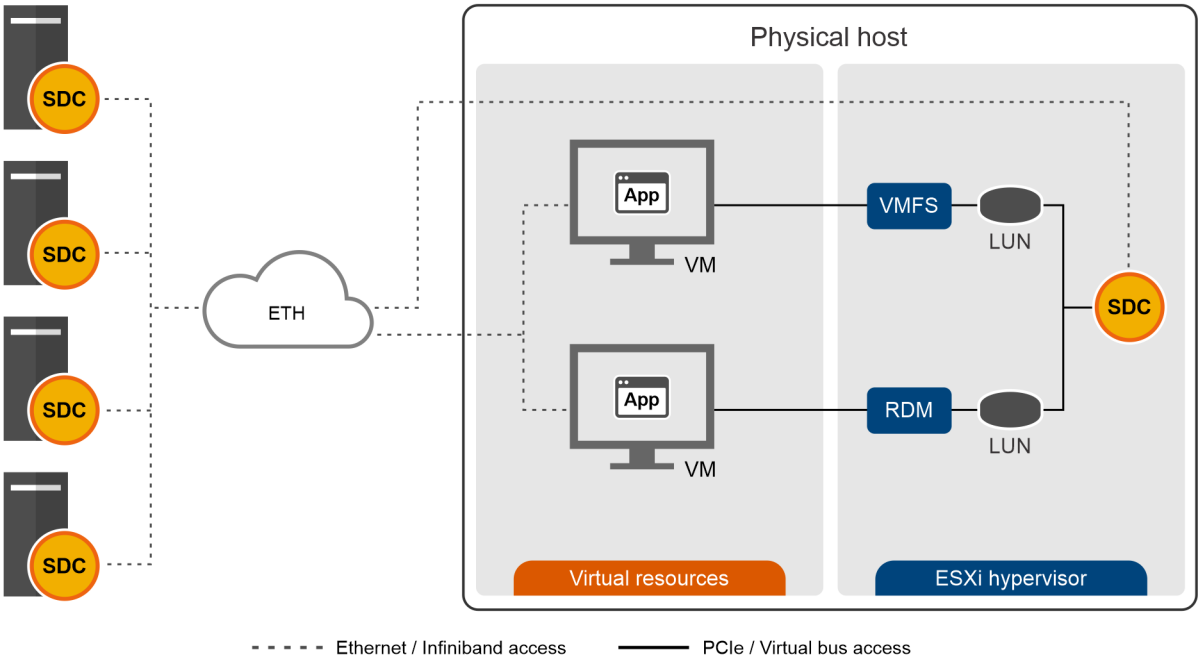


Figure 19. PowerFlex implementation in VMware ESXi with SDC in VMkernel

The LUNs in this example can be formatted with VMFS, and then exposed using the VMware ESXi host to the virtual machine, or the LUNs (volumes) that are NVMe target-based can be used as RDM devices. When the LUNs are used as RDM devices, the VMFS layer is omitted.

Supported VMware vSphere features

PowerFlex supports the following VMware vSphere features:

- vMotion
- Storage vMotion
- Fault tolerance
- DRS
- Storage DRS
- VAAI (except full copy primitive)

VMware ESXi vStorage APIs for Array Integration

VMware ESXi vStorage APIs for Array Integration provides hardware acceleration functionality. It allows the host to offload specific virtual machine and storage management operations to compliant storage hardware. With the assistance of the storage hardware, the host performs these operations faster, and consumes less CPU, memory, and storage fabric bandwidth.

VAAI uses these fundamental operations:

- Atomic Test & Set, which is used during creation and locking of files on the VMFS volume.
- Clone Blocks/Full Copy/XCOPY, which is used to copy or migrate data within the same physical array.
- Zero Blocks/Write Same, which is used to zero-out disk regions.
- Thin Provisioning in VMware ESXi 7.x and later hosts, which allows the VMware ESXi host to tell the array when the space previously occupied by a virtual machine (whether it is deleted or migrated to another datastore) can be reclaimed on thin provisioned LUNs.
- Block Delete in VMware ESXi 7.x and later hosts, which allows for space to be reclaimed using the SCSI UNMAP feature.

The PowerFlex supported VAAI features are:

- Atomic Test & Set
- Zero Blocks/Write Same
- Thin Provisioning in VMware ESXi 7.x and later hosts
- Block Delete in VMware ESXi 7.x and later hosts

The following is an output example:

```
esxcli storage core device vaai status get -d
eui.7dbf14034834bbe01bf7e55800000002
eui.7dbf14034834bbe01bf7e55800000002
VAAI Plugin Name:
ATS Status: supported
```

`Clone Status: unsupported` This means that Clone Block/Full Copy/Xcopy is not supported.

`Zero Status: supported` This means that write same is supported.

`Delete Status: supported` This means that UNMAP is supported.

NOTE: Thin provisioning is not shown in VAAI output.

VMware vSphere Storage I/O Control

While VMware Sphere Storage I/O Control (SIOC), Storage I/O Statistics Collection (SIOSC), and Network I/O Control (NIOC) are useful for vSan environments, their implementation may cause significant issues in a PowerFlex environment. Therefore, use of these options are not supported.

PowerFlex provides integrated capabilities to limit network bandwidth and IOPS limits for each volume for each storage data client.

Configuring direct attached storage

PowerFlex works with any free capacity: internal or direct-attached devices, and flash-based devices such as solid state drives (SSDs) and PCIe SSDs (NVMe drives). Although PowerFlex can work with any device topology, Dell Technologies recommends that you configure the raw devices as stand-alone devices.

Device data is erased when devices are added to a storage node. When adding a device to a storage node, PowerFlex checks that the device is clear before adding it. An error is returned per device if the device is found not to be clear. You can override this check by using the force device takeover option.

The following devices are considered to be not "clear," and cannot be added to a storage node:

- A complete device with either a file system or partition, or a partitioned device with a file system
- Devices in an LVM group
- Multipath devices (cannot be added as storage node devices)

The following limitations apply to direct attached storage:

- PowerFlex does not prevent SAN devices from being added.
- Devices in an LVM group cannot be added to a storage node.
- Within the database devices, only Oracle ASM devices can be detected and blocked.

NOTE:

If the server has a RAID controller, PowerFlex prefers to use the caching abilities of the controller for better performance. For best performance when using the controller, configure all devices as stand-alone by setting each of the devices separately to RAID 0 separately.

9

Maintenance

PowerFlex maintenance is limited primarily to configuration changes to the physical and virtual layers. It requires minimal user attention. When maintenance or planned restart of storage nodes is required, the maintenance mode feature can be used to streamline system operation.

Maintenance mode

Using maintenance mode, you can restart a storage node without requiring a shutdown or interrupting application I/Os.

When you place a storage node into maintenance mode, the node is immediately and temporarily removed from PowerFlex without building a new copy of the data. Throughout the duration of the maintenance mode, new writes are written to the devices on the other storage nodes in the cluster.

Degraded reads are used.

After the node is removed from maintenance mode, PowerFlex applies only the new writes to the node that was under maintenance, and the process is complete.

Maintain the physical layer

In the physical layer, maintenance is limited to adding and removing hardware units and configuring them into PowerFlex. These operations are usually a result of one of the following:

- Scaling out PowerFlex when there is a need for additional capacity by adding more storage devices to existing servers or adding additional servers.
- Device or server failure, where the storage or persistent memory device or the entire server needs to be replaced.

In both these cases, the maintenance operation requires you to add or remove storage capacity—and in some cases a storage node and its associated storage media—from the PowerFlex configuration. As far as PowerFlex is concerned, all these activities translate to storage node reconfigurations.

When adding or removing storage or persistent memory devices, you are required to add or remove the media from the device group with which it is associated. PowerFlex seamlessly redistributes the data stored on those devices accordingly.

If the server that was removed is a storage data client (SDC) node, or the server to be added requires exposing storage locally, SDC reconfiguration is also required.

Updates to the operating system of the server are an additional maintenance procedure that might affect the physical layer. In this case, performing a graceful shutdown or reboot operation returns the node to a fully operational state.

Maintaining the virtualization layer

You can perform the following maintenance operations on volumes exposed by the PowerFlex virtual SAN.

- Add or remove a volume by creating or deleting a volume in the system.
- Increase volume size by adding capacity to a given volume, as needed. The change in volume size occurs seamlessly without interrupting I/Os.
- Map or unmap volumes to an SDC or NVMe host. This enables or disables access to a volume by SDC or NVMe host—and by customer applications residing on the same node.
- Perform these MDM operations:
  - Move the MDM role to another role.
  - Add an MDM VIP.
  - Upgrade an MDM, PowerFlex data server, device gateway target, SDC, NVMe host, or LIA.

PowerFlex failures and recovery cases

The following table lists the different failure cases that might occur during operation of PowerFlex and the available recovery options:

Table 29. Common PowerFlex maintenance procedures

Scenario	Maintenance flow
Replace a failed disk (hardware failure) ● Not visible in the operating system ● No I/Os to disk	1. Identify the disk and node. 2. Remove the disk from the PowerFlex storage node. 3. Remove the disk from CloudLink encryption, if applicable. 4. Remove the disk from the RAID configuration, if applicable. 5. Blink the disk and remove it from the node. (In some cases this step might be performed after step 1.) 6. Physically add the new disk to the node. 7. Configure RAID in the RAID controller, if applicable. 8. Configure CloudLink, if applicable. 9. Add the disk to the PowerFlex storage node. In end of capacity situations,add the new disk first, and then remove the failed disk.
Replace a failed disk (logical failure) ● Due to I/O wait, no I/Os are written to disk ● Visible in the operating system	1. Identify the disk and node. 2. Try to clear the device error and let rebuild start. 3. If the error persists, replace the disk: a. Remove the disk from the PowerFlex storage node. b. Remove the disk from CloudLink encryption, if applicable. c. Remove the disk from the RAID configuration, if applicable. d. Blink the disk and remove it from the node. (In some cases this step might be performed after step 1.) e. Physically add the new disk to the node. f. Configure RAID in the RAID controller, if applicable. g. Configure CloudLink, if applicable. h. Add the disk to the PowerFlex storage node. In end of capacity situations,add the new disk first, and then remove the failed disk.
Replace a disk that is suspect and the administrator wants to replace it. ● Still servicing I/Os ● Visible in the operating system	1. Identify the disk and node. 2. Remove the disk from the PowerFlex storage node. 3. Remove the disk from CloudLink encryption, if applicable. 4. Remove the disk from the RAID configuration, if applicable. 5. Blink the disk and remove it from the node. (In some cases this step might be performed after step 1.) 6. Physically add the new disk to the node. 7. Configure RAID in the RAID controller, if applicable. 8. Configure CloudLink, if applicable. 9. Add the disk to the PowerFlex storage node. In end of capacity situations,add the new disk first, and then remove the failed disk.
The wrong disk from the same node was removed, and rebuild starts immediately. ● Hardware failure ● Not visible in the operating system ● No I/Os are written to disk	1. Register the serial number of the disk and the slot location. 2. Physically reinsert the disk. 3. Correlate the slot to the logical device path `/dev/sdxx` or `/dev/nvmexxx`. 4. Clear the device error in the storage node and check that rebuild starts (only for changes needed). 5. Replace the failed disk as in scenario 1, above.
Planned maintenance on a failed node. If hardware replacement requires removing disks, you can add the disks	1. Check that the system is in **Optimal** state and move the MDM to a different node, if required. 2. Identify the node.

Table 29. Common PowerFlex maintenance procedures (continued)

Scenario	Maintenance flow
back in any order. The PowerFlex storage node/protection domain remembers ownership. (PowerFlex signs the disks.)	3. Stage an upgrade of anything that is required, such as the operating system, firmware, and drivers. 4. Place the node into maintenance mode. 5. Power off or reboot the node. 6. Replace the hardware component, if relevant, 7. Power on the node. 8. Check that the status of all components is in **Optimal** state. 9. Exit maintenance mode. 10. Wait for the rebuild to finish.
Unplanned maintenance on a failed node that is recoverable. The node powers on automatically.	1. Upon crash of the node, the system starts a rebuild. 2. Alerts are raised that node components are disconnected. 3. The node comes back up, and component alerts are cleared. 4. The rebuild takes into account that the node is back, and only the rebuild changes.
Unplanned maintenance on a failed node that is not recoverable automatically due to a hardware failure	1. Upon crash of the node, the system starts a rebuild. 2. Alerts are raised that node components are disconnected. 3. The administrator sees that the node does not boot. Dell Technologies Support is engaged. 4. The boot issue is resolved, and power on continues. 5. The rebuild takes into account that the node is back, and only the rebuild changes.
Node failed due to operating system boot device corruption	1. Upon crash of the node, the system starts a rebuild. 2. Alerts are raised that node components are disconnected. 3. The administrator sees that the node does not boot due to an operating system issue, Dell Technologies Support is usually engaged. Option 1: Recover the node. 1. Reinstall the operating system of the node. Dell Technologies Support helps install components manually and reconfigure the storage node device IDs to match the original device IDs. 2. The component services are started after configuration is complete. 3. Rebuild takes in to account that the node is back, and only the rebuild changes. Option 2: Remove and add the node 1. Reinstall the operating system of the node. Dell Technologies Support helps remove the node and add the "new" node back for a full rebuild. 2. In end of capacity situations, add the new disk first, and then remove the failed disk.
Complete node failure. Node replacement is required.	1. Upon crash of the node, the system starts a rebuild. 2. Alerts are raised that node components are disconnected. 3. The administrator sees that the node does not boot due to an operating system issue , Dell Technologies Support is usually engaged. 4. Option 1 : Remove the failed node and add a replacement node. 5. Option 2: Reinstall the operating system of the node. Have Dell Technologies Support help remove the node and add the "new" node back for a full rebuild.
PMEM failure	1. Multiple PMEM devices fail. 2. Remove the PMEM devices from the device group. 3. Place the node into maintenance mode. 4. Power off the node. 5. Replace the PMEM devices and sanitize them, if necessary. 6. Power on the node. 7. Add the PMEMs to the device group.

Table 29. Common PowerFlex maintenance procedures (continued)

Scenario	Maintenance flow
	Rebuild /rebalance starts and the system returns to **Optimal** state upon completion.
Network failure	Scenario 1: One network out of two or more has failed. 1. The system continues to work over a single or the remaining networks. 2. When the network recovers, network high availability and load balancing resume. Scenario 2: The entire system network fails. 1. Data is unavailable in the system, contact Dell Technologies Support. 2. If needed, place protection domains in **Inactive** state to avoid a rebuild when the system comes back online. 3. When the network recovers, the system returns to operational state. Activate the protection domains, if required.
Unintended full shutdown due to power loss	1. System was unresponsive or is down due to power loss. 2. Power on all nodes. You can power on the MDM either first or last to allow placing some protection domains in **Inactive** state. 3. Check that all components are connected and that the protection domains and storage pools are on line and in **Optimal** state. a. To avoid a rebuild, optionally place some protection domains in **Inactive** state. b. When all components are back on line, activate the protection domains. The system should come back up without the need to rebuild.
Intended full shutdown	1. Stop the I/Os. 2. Place all protection domains in **Inactive** state. 3. Shut down all nodes. 4. After some time, power on all nodes. 5. Check that all nodes are connected to the MDM and are in **Optimal** state. 6. Activate all protection domains. No rebuild occurs. 7. The system is ready to resume I/Os.
Storage pool is out of capacity	Option 1: 1. A storage pool has reached end of capacity. 2. I/Os stop, but are still available to perform reads. 3. Dell Technologies Support helps to move volumes to other storage pools and recover from end of capacity in storage pools. Option 2: 1. A storage pool reaches end of capacity. 2. I/Os stop, but are still available to perform reads. 3. Resize the storage pool or add a node or storage disks to the storage pool. A rebalance starts. 4. Resume I/Os once free capacity is available.
NDU failure	During a nondisruptive upgrade (NDU), a storage node fails to start after the upgrade. **NOTE:** The system is still working because the storage node is in maintenance mode. Option 1: 1. Restore the storage node to the previous version and restart it. 2. Exit maintenance mode and wait for the rebuild to finish. 3. Fix any issues and restart the upgrade. **NOTE:** The node might be in mid-upgrade state for weeks. Option 2:

Table 29. Common PowerFlex maintenance procedures (continued)

Scenario	Maintenance flow
	1. Fix the issue that prevents the storage node from restarting. (Perhaps the problem is an operating system issue or a lack of resources.) 2. If necessary, exit maintenance mode and wait for the rebuild to finish. 3. Retry the upgrade and go to the next step in the process, according to whether the upgrade is automated or manual.
SDC - MDM version mismatch	No such case can occur. The minimum SDC version is 5.0, and starting with PowerFlex 5.0, the SDCs can connect to any version of the MDM and are only able to use the features supported by both sides (the SDC and backend).
Node or system secure erase of the disks and data	Use an external tool to scrub the data and node.

Monitoring

PowerFlex supports a number of software and hardware monitoring features to assist you with system maintenance.

SDC connection monitoring

The system monitors all connections between SDCs and storage nodes and sends out an alert when an active connection between an SDC and a storage node goes down.

To monitor SDC and storage node connections effectively, the MDM collects connectivity updates from all the SDCs. The MDM posts events whenever an SDC connects to or disconnects from a specific storage node IP address.

The MDM frequently analyzes the connectivity status to determine the current system state. The system does not send out alerts for temporary connectivity issues that are resolved in less than 10 seconds.

The following are the possible connectivity states between SDCs and storage nodes in the system:

- All connected (SDCs are connected to the backend storage components.)
- One SDC is disconnected from one storage node.
- One SDC is disconnected from one storage node IP address.
- One SDC is disconnected from all storage nodes.
- All SDCs are disconnected from one storage node.
- All SDCs are disconnected from one storage node IP address.
- All SDCs are disconnected from all storage nodes.
- Multiple disconnections (Multiple SDCs are disconnected from multiple backend storage components).

NOTE: The storage node contains the PowerFlex data server (PDS) and device gateway target (DGWT) components.

When the connectivity state of the system changes to any state other than *All Connected*, an alert is displayed in PowerFlex Manager and is written to the MDM event log. Once an alert is generated, you can use the SCLI to query details on the disconnection using the following command:

```
scli --query_sdc_to_storage node_disconnections
```

The MDM does not monitor the connectivity state of SDCs or storage nodes in the following scenarios:

- The storage node is in maintenance mode.
- The storage node is disconnected from the MDM.
- The storage node is in the process of being removed.
- The SDC is disconnected from the MDM for more than two minutes.
- The SDC is not approved.

SMART hardware monitoring

The PowerFlex bare-metal solution provides monitoring capabilities for RAID controllers and storage devices compatible with SMART (Self-Monitoring, Analysis and Reporting Technology) protocols.

In Linux-based environments, SMART-compatible SSDs and RAID storage controllers can be monitored for SMART attributes such as temperature and SSD wear level. LEDs can also be lit on these hardware devices, to simplify physical identification for maintenance purposes.

Each hardware vendor defines specific thresholds for the SMART attributes. This feature currently supports storage devices controlled by LSI, HP, and Dell RAID controllers, and standalone devices. During system deployment, an external monitoring tool is installed as part of the LIA on each node. Additional RAID controller tools must be installed manually after system deployment: StorCLI for LSI RAID controllers, HPSSACLI for HP RAID controllers, or PERCCLI for Dell RAID controllers. The system uses these tools to collect the counters that are returned to the MDM.

NOTE: Sometimes, LSI RAID controllers may report vendor information as "AVAGO" instead of LSI.

The MDM queries the storage nodes at set intervals, and stores the returned information. This information can be viewed using CLI queries. In addition, when thresholds are crossed for SMART attributes, the system generates alerts.

When the CLI is used to query device information, physical device information, such as serial number, disk slot information, model name, vendor, temperature, and wear level information (for SSDs only) is included in the returned response.

You can use PowerFlex Manager to monitor SMART-related alerts in the **Alerts** view.

In addition, SNMP traps and secure connect gateway alert codes can be used to monitor alerts triggered by devices compatible with SMART.